# Introduction to GenerativeAI and Retrieval-Augmented Generation (RAG)


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2. Python working environment - Either one of the options below 2
3. Large Language Model(LLM) Model - Either one of the options below 2
4. Integrated Development Environment (IDE) - Either one of the options below 3

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**Introduction to Retrieval-Augmented Generation (RAG)** **51**

# Prerequisites

1. Python Distribution - Either one of the options below
    a. Open source or classic Python(from https://www.python.org/downloads/ )
    b. Anaconda Python(from https://www.anaconda.com/download)

2. Python working environment - Either one of the options below
    a. Python virtual environment
    b. Python root installation

3. Large Language Model(LLM) Model - Either one of the options below
    a. LLM API Key(Paid or Free)
        i. **LLM Providers**
            1. OpenAI's GPT model family
            2. Anthropic's Claude model family
            3. Facebook/Meta's Llama model family
            4. Google's Gemini model family

Create account and Get API key
Url: https://aistudio.google.com/

            5. Alibaba Cloud's Qwen model family
            6. DeepSeek model family
            7. Z.ai's GLM model family
            8. Kimi model family
            9. Twitter/X's Grok model family
            10. NVIDIA's Nemotron model family

        ii. **LLM Inferencing Providers/Hosting Providers**
            1. Hugging Face

Create account and Get API key
Url: https://huggingface.co/

            2. Groq

Create account and Get API key
Url: https://groq.com/

            3. Ollama Cloud

        iii. **LLM API Resellers**
            1. OpenRouter

Create account and Get API key
Url: https://openrouter.ai/

            2. APInex

Create account and Get API key
Url: https://apinex.bond/

iv. **LLM Gateway**
    1. LiteLLM
    2. Bifrost
    3. OmniRoute (self-hosting option available)
    4. CLIProxyAPI (self-hosting option available)

**b. Local LLM Model Setup**

i. Using LLM Model file with Transformer python package
**ii. Ollama**
iii. LMStudio
iv. vLLM

4. Integrated Development Environment (IDE) - Either one of the options below

a. Visual Source Code

IDE Software Download Url: https://code.visualstudio.com/

b. PyCharm

IDE Software Download Url: https://www.jetbrains.com/pycharm/

c. Jupyter Notebook/Lab
d. Google Colab

Environment Url: https://colab.research.google.com/

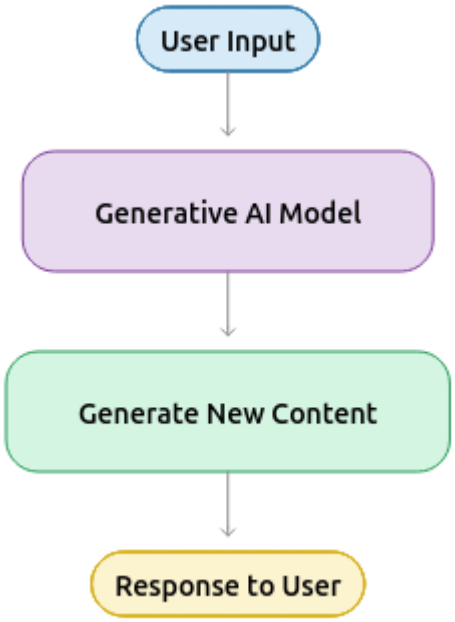
e. Python CLI(Command-Line Interface) / REPL(Read–Eval–Print Loop)

# Getting Started

Welcome to the world of Generative AI!

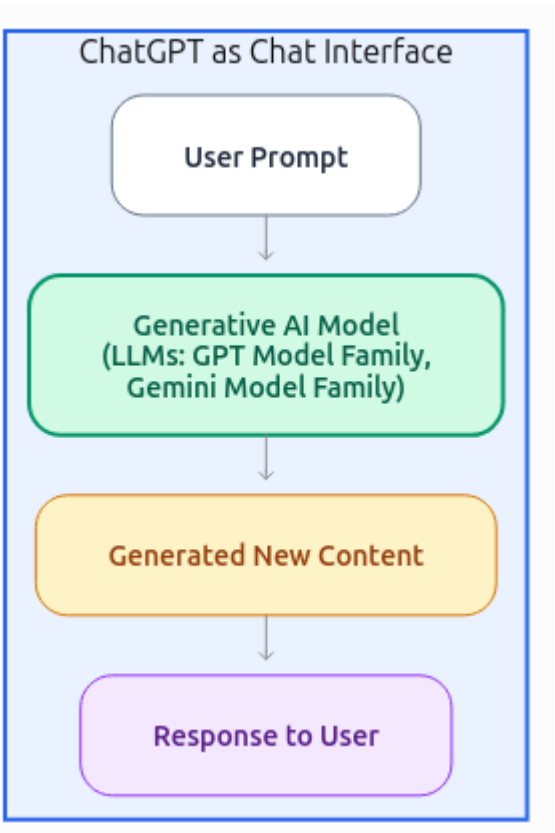
## Generative AI(GenAI)

**Generative AI** is a branch of artificial intelligence that creates new content—such as text, images, audio, video, code, and 3D designs—in response to user prompts.




**Example:**

1. **User Input:** "Write a poem about AI."
2. **Generative AI Model:** Understands the **prompt** and generates content.
3. **Generate New Content:** Creates a poem.
4. **Response to User:** Displays the poem.




ChatGPT: https://chatgpt.com/

Depending on the model, the generated content can include:

- **Text:** Answers, emails, reports, summaries, and stories.
- **Images:** Illustrations, logos, artwork, and product designs.
- **Audio:** Speech, music, and sound effects.
- **Video:** Video clips, animations, and visual scenes.
- **Code:** Python programs, SQL queries, and application components.
- **Structured data:** JSON objects, tables, and extracted business information.

**Slowly we are Stepping into the World of Generative AI Concepts/Terminologies**

1. LLM
2. Prompt
3. Token
4. API Key

Large Language Models (LLMs) & Core AI Concepts

• LLM (Large Language Model): An AI neural network trained on massive datasets of text to understand, generate, and predict human-like language. Most modern LLMs use the **Transformer architecture** to predict and output text one piece at a time based on statistical probabilities.

• Prompt: The text input or instruction provided by a user to guide the LLM's response. The wording, context, and structure of a prompt directly influence the quality and format of the model's output.

• Token: The basic unit of text that an LLM processes. Instead of reading full words, a tokenizer chops text into chunks. In English, a token is roughly 4 characters or 0.75 words; common words might be one token, while rare words are split into multiple tokens. LLMs have a strict **context window** (total token capacity) and are typically priced per token.

• API Key: A unique, secure password used by developers to authenticate their code when connecting to a hosted LLM. It tracks usage, handles billing, and should always be kept secure in server-side environment variables rather than hardcoded into applications.

**User prompt:** Explain generative AI in 3 simple sentences.

## How to connect Google's Gemini Model:

A simple step-by-step guide to calling Google's Gemini LLM from Python.

## Step 1: Install Python

The Gemini library needs Python 3.9 or newer. Check your version:

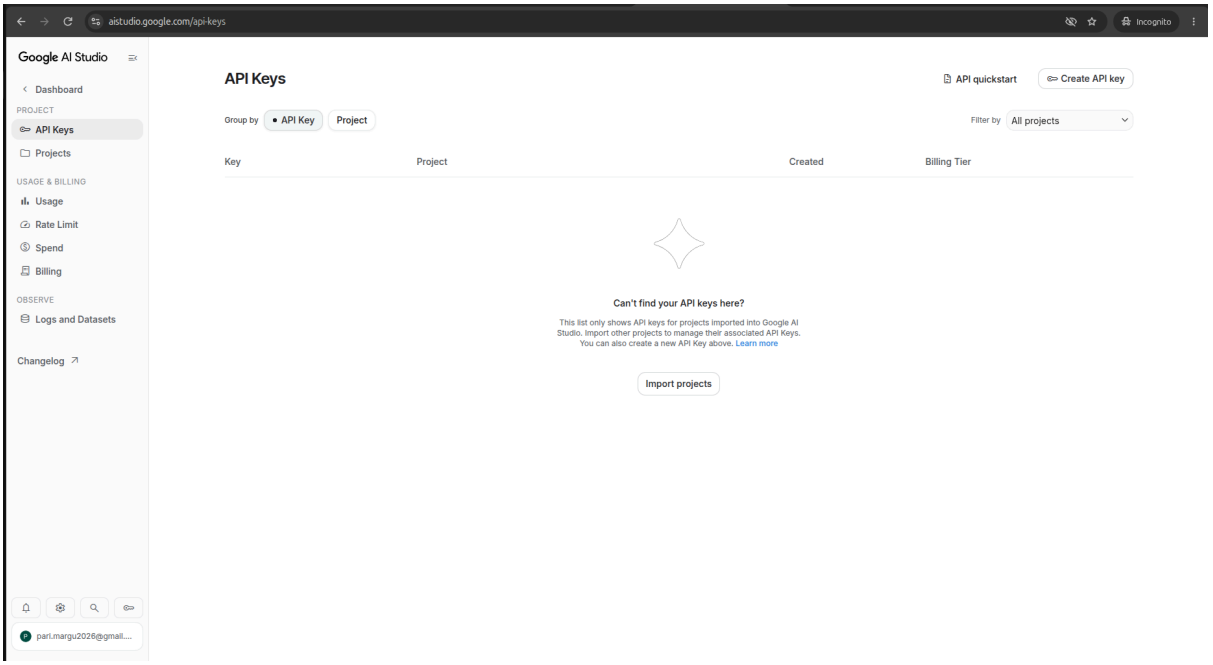
python --version

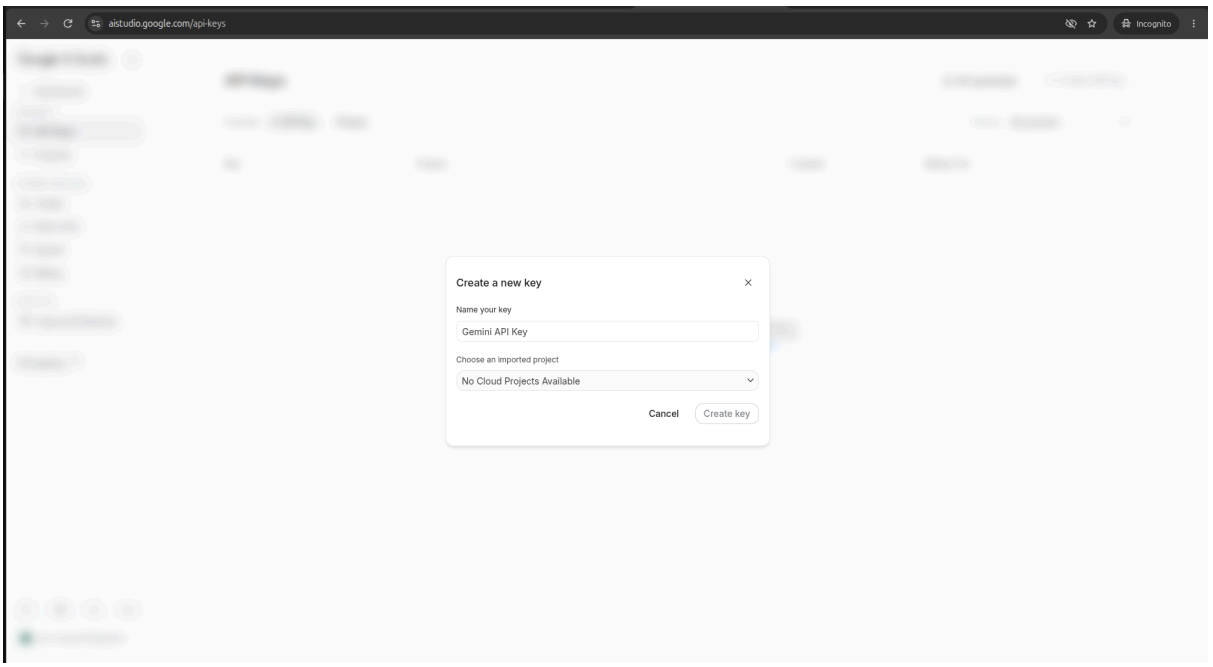
## Step 2: Get a Gemini API key

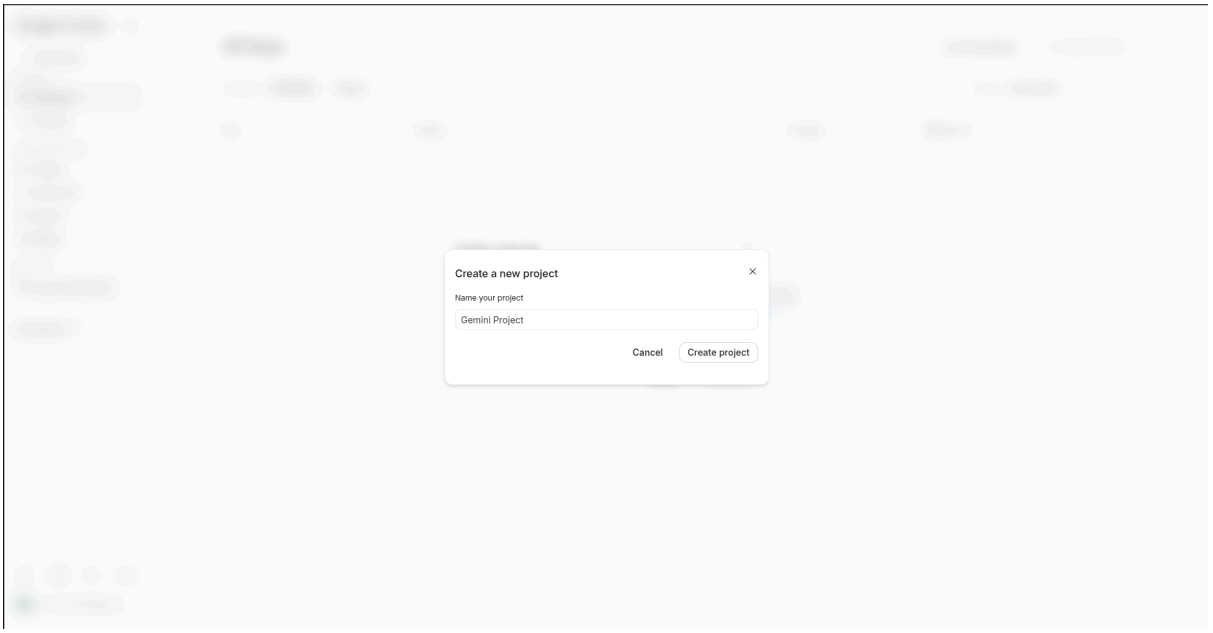
1. Go to **aistudio.google.com** and sign in with your Google account.
2. Click **Get API key**, then **Create API key**.
3. Copy the key and keep it private.

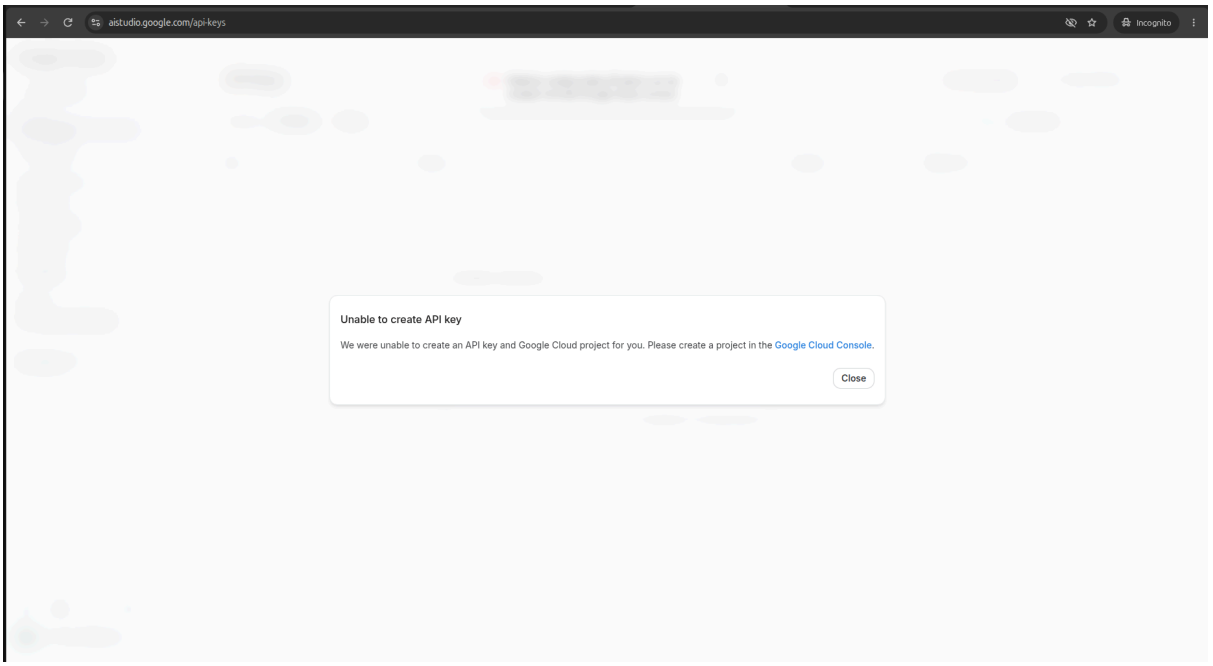
Google AI Studio offers a free tier, so you can usually start testing without adding billing.

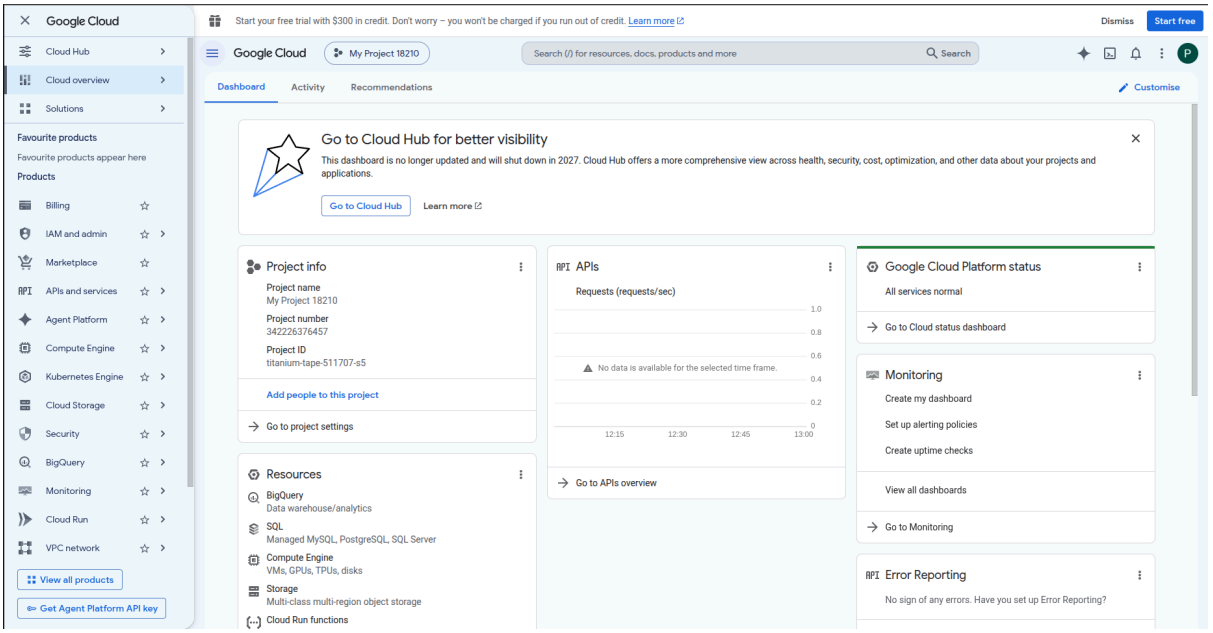
https://aistudio.google.com/api-keys

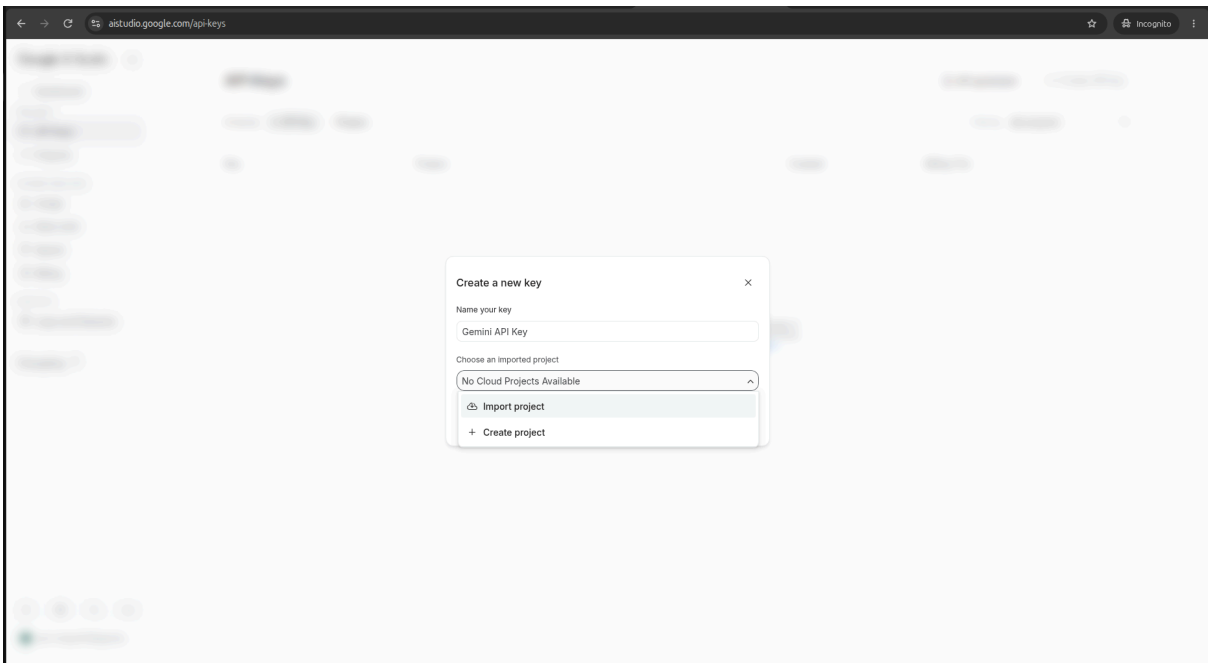

aistudio.google.com/api-keys
Incognito
Google AI Studio
Dashboard
PROJECT
API Keys
Projects
USAGE & BILLING
Usage
Rate Limit
Spend
Billing
OBSERVE
Logs and Datasets
Changelog
pari.margu2026@gmail....
API Keys
API quickstart
Create API key
Group by
API Key
Project
Filter by
All projects
Key
Project
Created
Billing Tier
Can't find your API keys here?
This list only shows API keys for projects imported into Google AI Studio. Import other projects to manage their associated API Keys.
You can also create a new API Key above. Learn more
Import projects

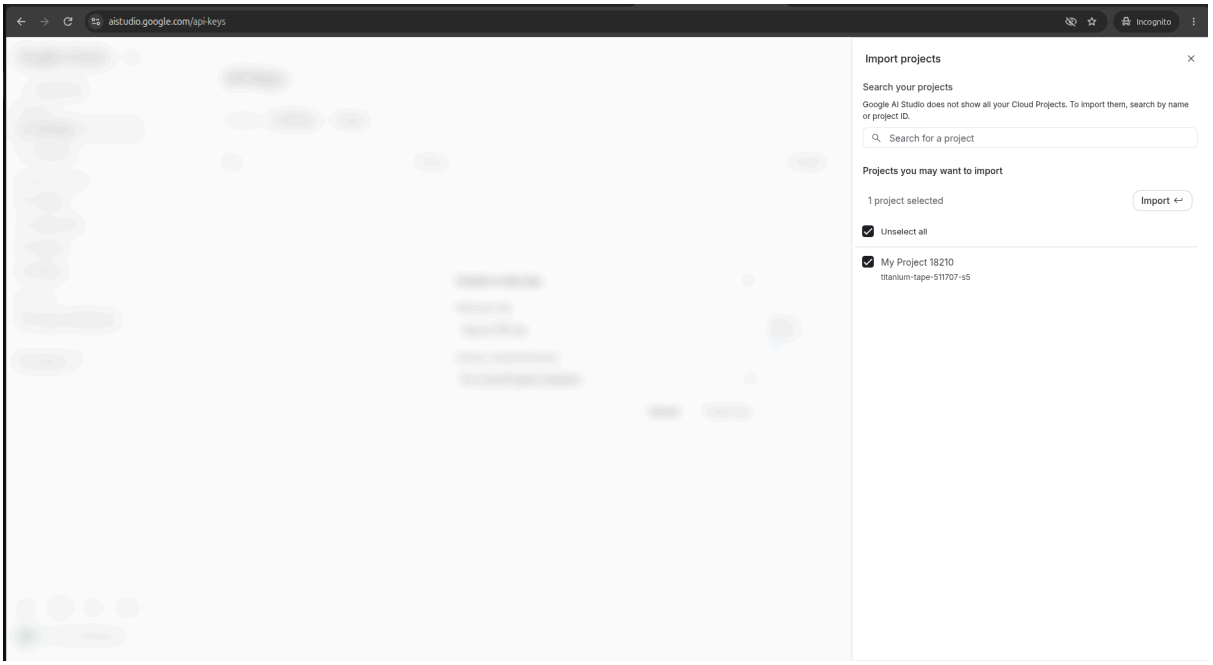

aistudio.google.com/api-keys
Incognito
Create a new key
Name your key
Gemini API Key
Choose an imported project
No Cloud Projects Available
Cancel
Create key

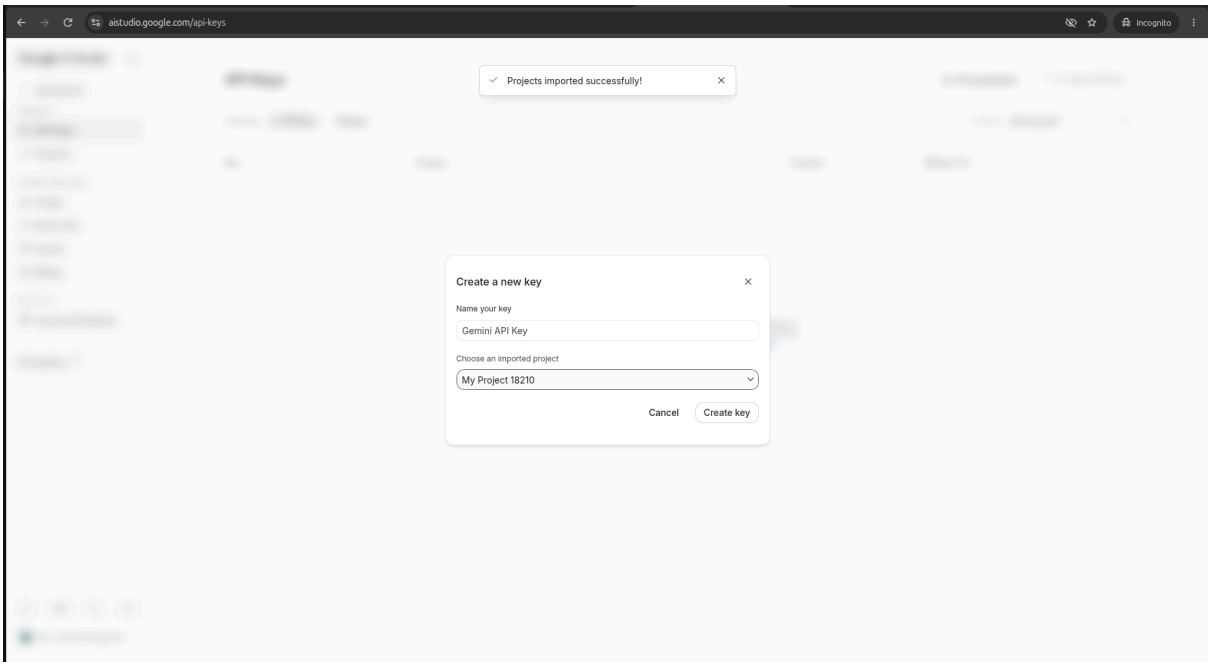

Create a new project
Name your project
Gemini Project
Cancel
Create project

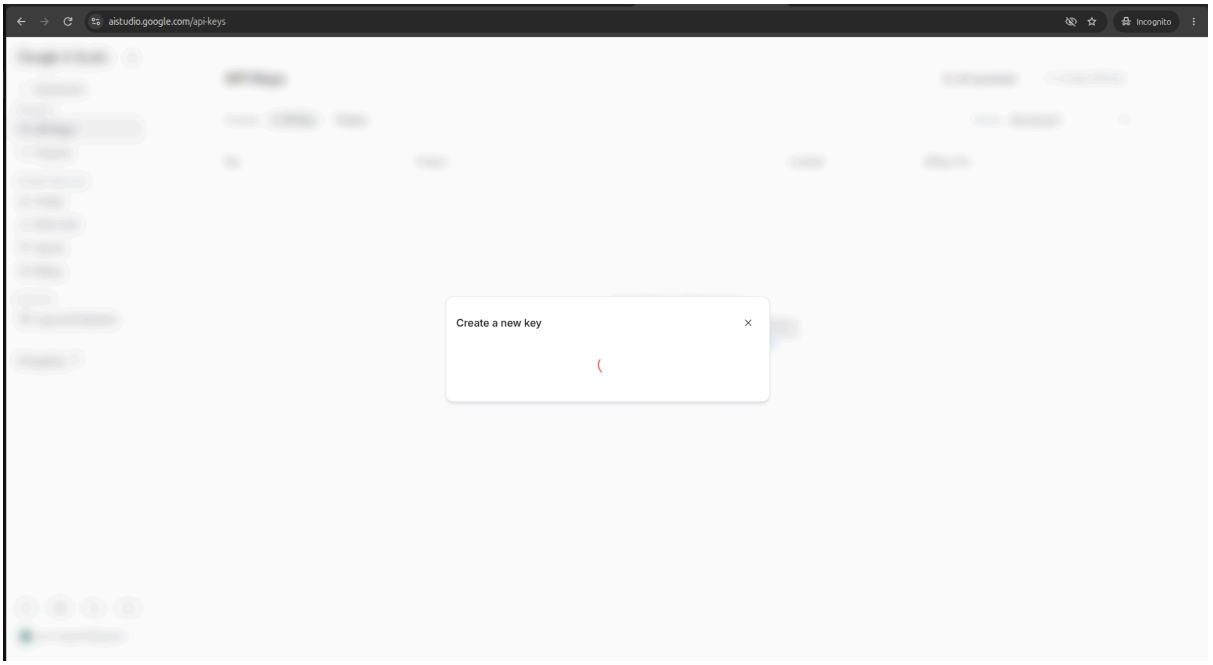

aistudio.google.com/api-keys
Incognito
Unable to create API key
We were unable to create an API key and Google Cloud project for you. Please create a project in the Google Cloud Console.
Close

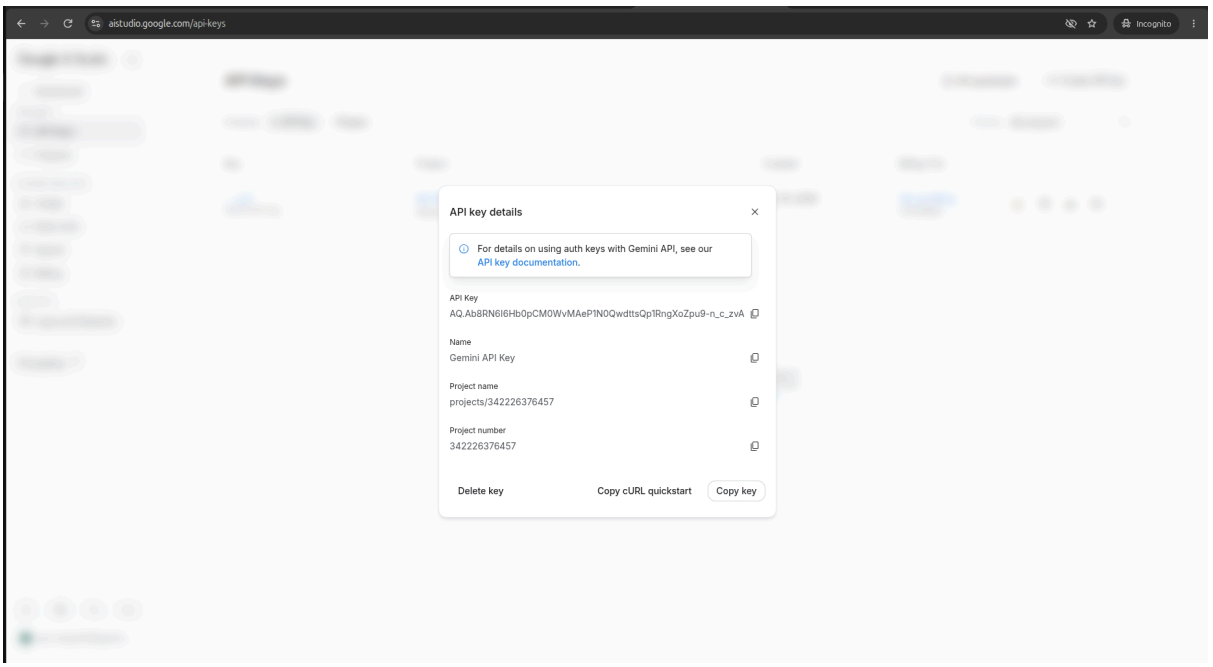

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Start your free trial with $300 in credit. Don't worry – you won't be charged if you run out of credit. Learn more
Dismiss
Start free
Google Cloud
My Project 18210
Search (/) for resources, docs, products and more
Search
Cloud Hub
Cloud overview
Solutions
Favourite products
Favourite products appear here
Products
Billing
IAM and admin
Marketplace
APIs and services
Agent Platform
Compute Engine
Kubernetes Engine
Cloud Storage
Security
BigQuery
Monitoring
Cloud Run
VPC network
View all products
Get Agent Platform API key
Dashboard
Activity
Recommendations
Customise
Go to Cloud Hub for better visibility
This dashboard is no longer updated and will shut down in 2027. Cloud Hub offers a more comprehensive view across health, security, cost, optimization, and other data about your projects and applications.
Go to Cloud Hub
Learn more
Project info
Project name
My Project 18210
Project number
342226376457
Project ID
titanium-tape-511707-s5
Add people to this project
Go to project settings
APIs
Requests (requests/sec)
No data is available for the selected time frame.
12:15
12:30
12:45
13:00
Go to APIs overview
Google Cloud Platform status
All services normal
Go to Cloud status dashboard
Monitoring
Create my dashboard
Set up alerting policies
Create uptime checks
View all dashboards
Go to Monitoring
Error Reporting
No sign of any errors. Have you set up Error Reporting?
Resources
BigQuery
Data warehouse/analytics
SQL
Managed MySQL, PostgreSQL, SQL Server
Compute Engine
VMs, GPUs, TPUs, disks
Storage
Multi-class multi-region object storage
Cloud Run functions


aistudio.google.com/api-keys
Incognito
Create a new key
Name your key
Gemini API Key
Choose an imported project
No Cloud Projects Available
Import project
Create project


aistudio.google.com/api-keys
Incognito
Import projects
Search your projects
Google AI Studio does not show all your Cloud Projects. To import them, search by name or project ID.
Search for a project
Projects you may want to import
1 project selected
Import
Unselect all
My Project 18210
titanium-tape-511707-s5

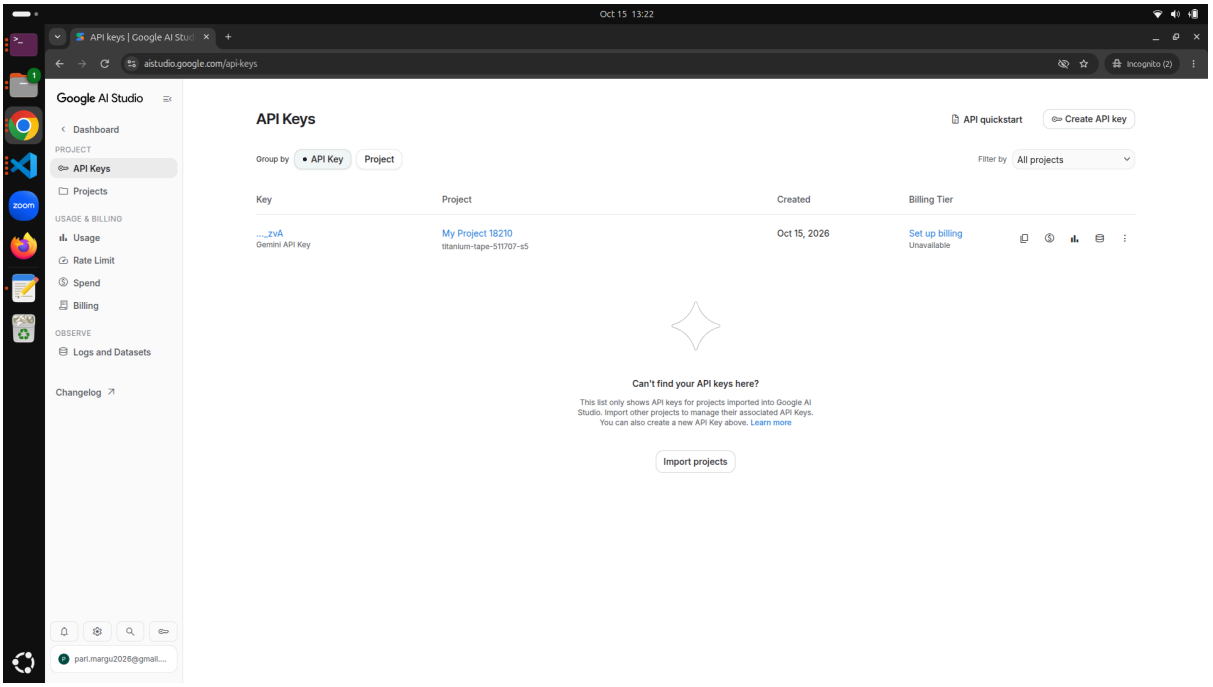

aistudio.google.com/api-keys
Incognito
Projects imported successfully!
Create a new key
Name your key
Gemini API Key
Choose an imported project
My Project 18210
Cancel
Create key

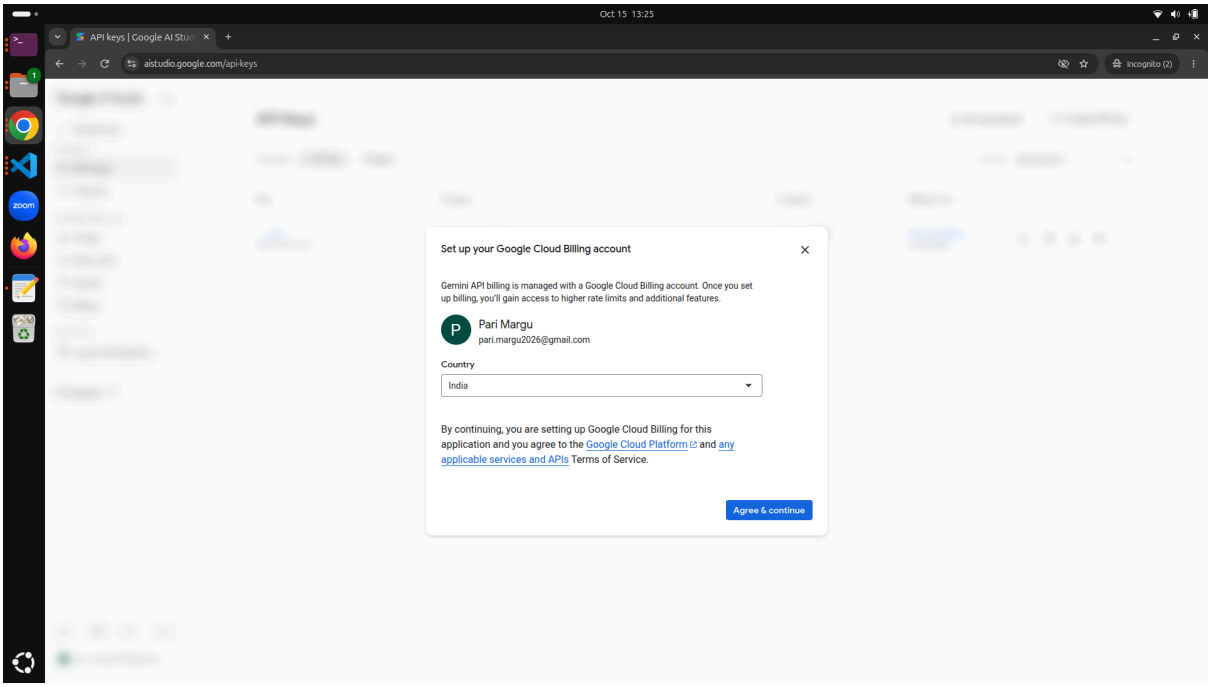


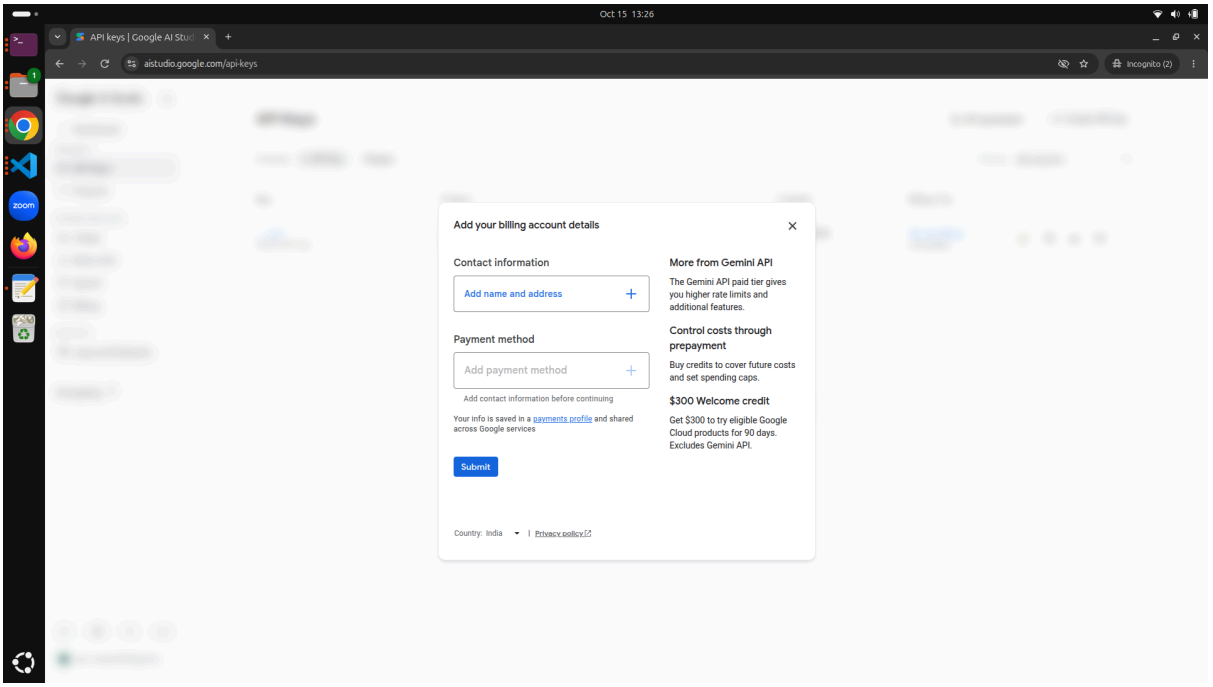



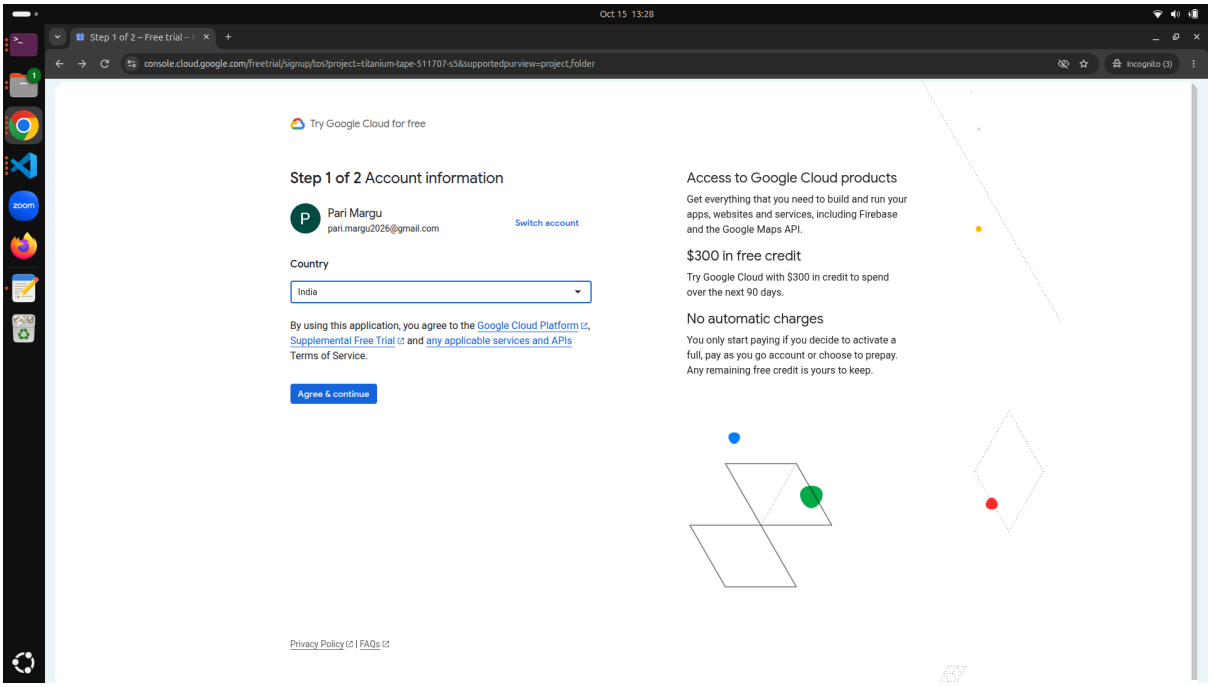

If you face the below issue,

```
ClientError: 403 PERMISSION_DENIED. {'error': {'code': 403, 'message': 'Your project has been denied access. Please contact support.', 'status': 'PERMISSION_DENIED'}}
```


Oct 15 13:22
API keys | Google AI Stud
aistudio.google.com/api-keys
Incognito (2)
Google AI Studio
Dashboard
PROJECT
API Keys
Projects
USAGE & BILLING
Usage
Rate Limit
Spend
Billing
OBSERVE
Logs and Datasets
Changelog
pari.margu2026@gmail....
API Keys
API quickstart
Create API key
Group by
API Key
Project
Filter by
All projects
Key
Project
Created
Billing Tier
...zvA
Gemini API Key
My Project 18210
titanium-tape-511707-e5
Oct 15, 2026
Set up billing
Unavailable
Can't find your API keys here?
This list only shows API keys for projects imported into Google AI Studio. Import other projects to manage their associated API Keys.
You can also create a new API Key above. Learn more
Import projects


Oct 15 13:25
API keys | Google AI Stud
aistudio.google.com/api-keys
Incognito (2)
Set up your Google Cloud Billing account
Gemini API billing is managed with a Google Cloud Billing account. Once you set up billing, you'll gain access to higher rate limits and additional features.
P
Pari Margu
pari.margu2026@gmail.com
Country
India
By continuing, you are setting up Google Cloud Billing for this application and you agree to the Google Cloud Platform and any applicable services and APIs Terms of Service.
Agree & continue


Oct 15 13:26
API keys | Google AI Stud
aistudio.google.com/api-keys
Incognito (2)
Add your billing account details
Contact information
Add name and address
Payment method
Add payment method
Add contact information before continuing
Your info is saved in a payments profile and shared across Google services
Submit
Country: India
Privacy policy
More from Gemini API
The Gemini API paid tier gives you higher rate limits and additional features.
Control costs through prepayment
Buy credits to cover future costs and set spending caps.
$300 Welcome credit
Get $300 to try eligible Google Cloud products for 90 days. Excludes Gemini API.


Oct 15 13:28
Step 1 of 2 – Free trial
console.cloud.google.com/freetrial/signup/tos?project=titanium-tape-511707-s5&supportedpurview=project,folder
Incognito (3)
Try Google Cloud for free
Step 1 of 2 Account information
Pari Margu
pari.margu2026@gmail.com
Switch account
Country
India
By using this application, you agree to the Google Cloud Platform, Supplemental Free Trial and any applicable services and APIs Terms of Service.
Agree & continue
Privacy Policy | FAQs
Access to Google Cloud products
Get everything that you need to build and run your apps, websites and services, including Firebase and the Google Maps API.
$300 in free credit
Try Google Cloud with $300 in credit to spend over the next 90 days.
No automatic charges
You only start paying if you decide to activate a full, pay as you go account or choose to prepay. Any remaining free credit is yours to keep.

## Step 3: Install the Google GenAI library

pip install -U google-genai

The older package `google-generativeai` is outdated, so use `google-genai`.

## Step 4: Save the API key as an environment variable

This keeps the key out of your code.

**Windows (Command Prompt):**

setx GEMINI_API_KEY "your-api-key-here"

Close the Command Prompt and open a new one after running this.

**Mac / Linux:**

export GEMINI_API_KEY="your-api-key-here"

## Step 5: Write the Python code

Create a file named `app.py`:

```
from google import genai
from google.genai import types

# The client reads GEMINI_API_KEY from your environment automatically
client = genai.Client()

# 1. User prompt
user_prompt = "Explain generative AI in 3 simple sentences."

# 2. Send the prompt to the Gemini model
response = client.models.generate_content(
    model="gemini-3.5-flash",
    config=types.GenerateContentConfig(
        system_instruction="You are a helpful teacher who explains things simply."
    ),
    contents=user_prompt
)

# 3 & 4. Get the generated content and show it to the user
print(response.text)
```

## Step 6: Run it

python app.py

Gemini's answer will print in your terminal.

## How it matches previous GenAI's simple flow diagram

- **User prompt:** `user_prompt`
- **Generative AI model:** `client.models.generate_content(model="gemini-3.5-flash", ...)`
- **Generated new content:** `response.text`
- **Response to user:** `print(...)`

## **Bonus:** a simple chat loop

Gemini has a built-in chat feature that remembers earlier messages in the conversation:

```
from google import genai

client = genai.Client()
chat = client.chats.create(model="gemini-3.5-flash")

while True:
    user_prompt = input("You: ")
    if user_prompt.lower() == "exit":
        break

    response = chat.send_message(user_prompt)
    print("Gemini:", response.text, "\n")
```

Model names change over time, so check Google's Gemini models page for the latest names. If you see an authentication error, the API key isn't set correctly.

## How to connect OpenAI's GPT Model:

A simple step-by-step guide to calling an OpenAI LLM from Python.

## Step 1: Install Python

Check that Python 3.8 or newer is installed:

python --version

## Step 2: Get an OpenAI API key

1. Go to **platform.openai.com** and sign in.
2. Open **API keys** and click **Create new secret key**.
3. Copy the key and keep it private. It is shown only once.
4. Add billing or credits to your account, because the API is paid separately from a ChatGPT subscription.

## Step 3: Install the OpenAI library

pip install openai

## Step 4: Save the API key as an environment variable

This keeps the key out of your code.

**Windows (Command Prompt):**

setx OPENAI_API_KEY "your-api-key-here"

Close the Command Prompt and open a new one after running this.

**Mac / Linux:**

export OPENAI_API_KEY="your-api-key-here"

## Step 5: Write the Python code

Create a file named `app.py`:

```
from openai import OpenAI

# The client reads OPENAI_API_KEY from your environment automatically
client = OpenAI()

# 1. User prompt
user_prompt = "Explain generative AI in 3 simple sentences."

# 2. Send the prompt to the LLM
response = client.responses.create(
    model="gpt-5",
    instructions="You are a helpful teacher who explains things simply.",
    input=user_prompt
)
```

```
# 3 & 4. Get the generated content and show it to the user
print(response.output_text)
```

## Step 6: Run it

```
python app.py
```

The AI's answer will print in your terminal.

## How it matches previous GenAI's simple flow diagram

- **User prompt:** `user_prompt`
- **Generative AI model:** `client.responses.create(model="gpt-5", ...)`
- **Generated new content:** `response.output_text`
- **Response to user:** `print(...)`

## **Bonus:** a simple chat loop

This version keeps asking for questions until you type `exit`:

```
from openai import OpenAI

client = OpenAI()

while True:
    user_prompt = input("You: ")
    if user_prompt.lower() == "exit":
        break

    response = client.responses.create(
        model="gpt-5",
        input=user_prompt
    )
    print("AI:", response.output_text, "\n")
```

The `responses.create()` method is the API style OpenAI recommends for new projects. Model names change over time, so check OpenAI's models page for the latest names. If you see an authentication error, the API key isn't set correctly. If you see a quota error, you need to add credits to your account.

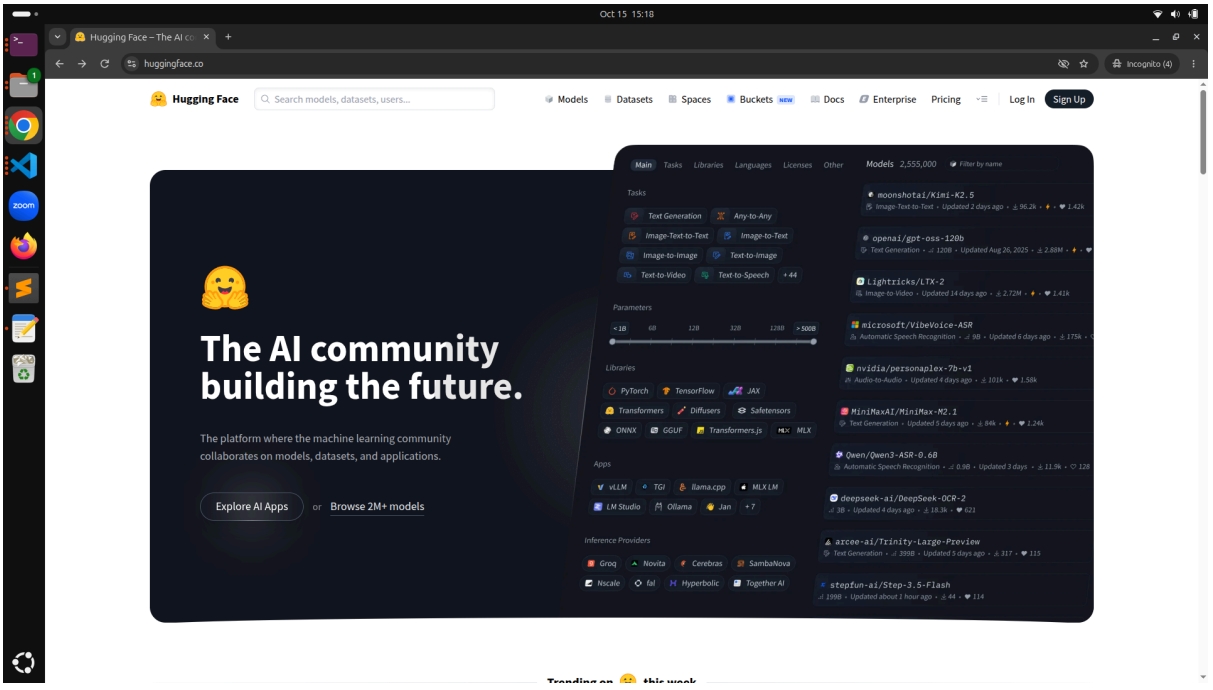
### OpenAI vs Gemini code at a glance

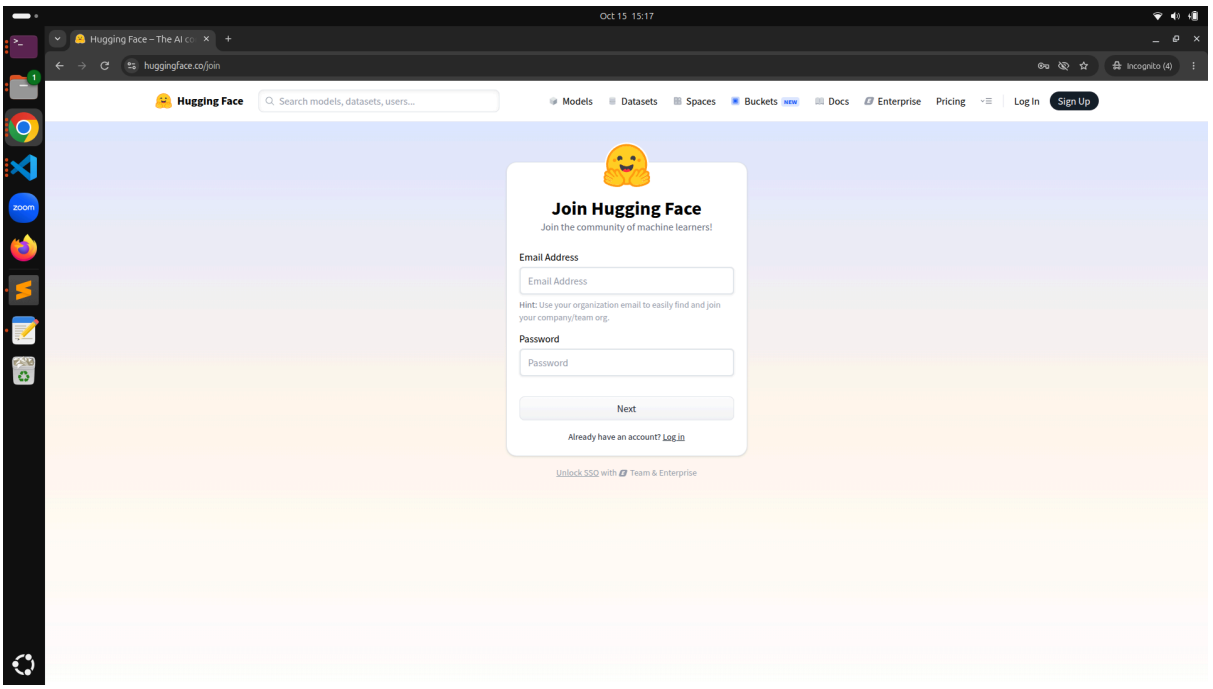
| Step | OpenAI | Gemini |
|---|---|---|
| Install | `pip install openai` | `pip install google-genai` |
| API key variable | `OPENAI_API_KEY` | `GEMINI_API_KEY` |
| Create client | `OpenAI()` | `genai.Client()` |
| Send prompt | `client.responses.create(...)` | `client.models.generate_content(...)` |
| Get answer | `response.output_text` | `response.text` |

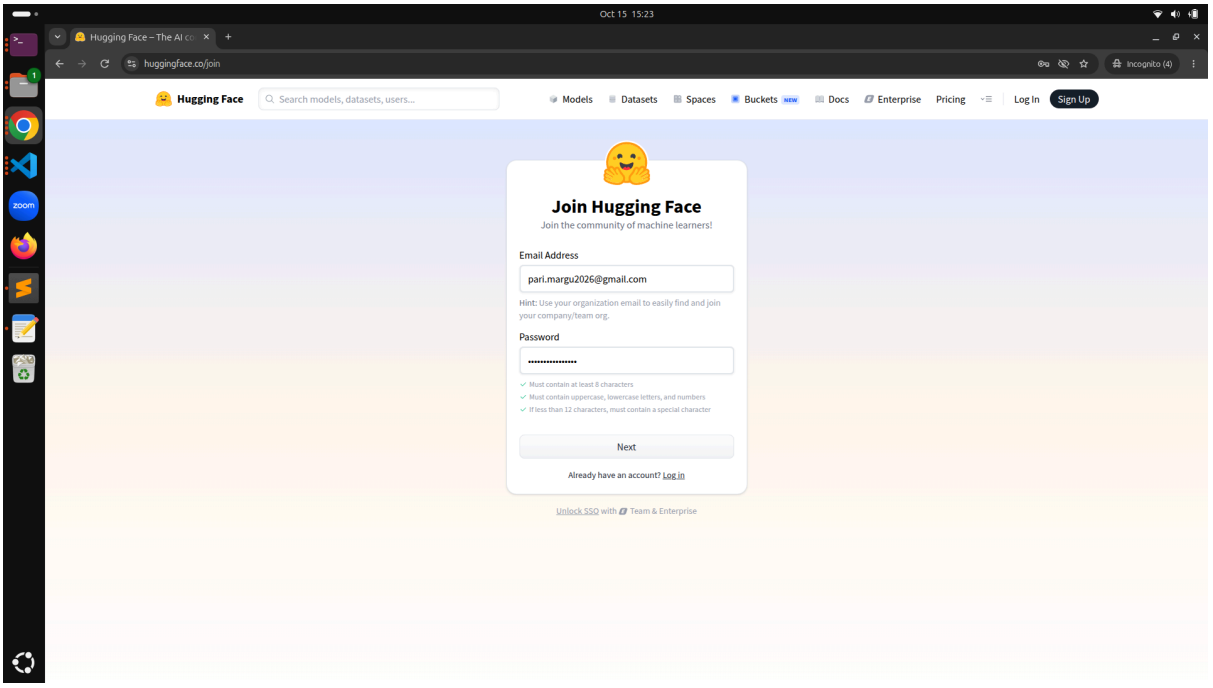
## How to connect Hugging Face's LLM Model:

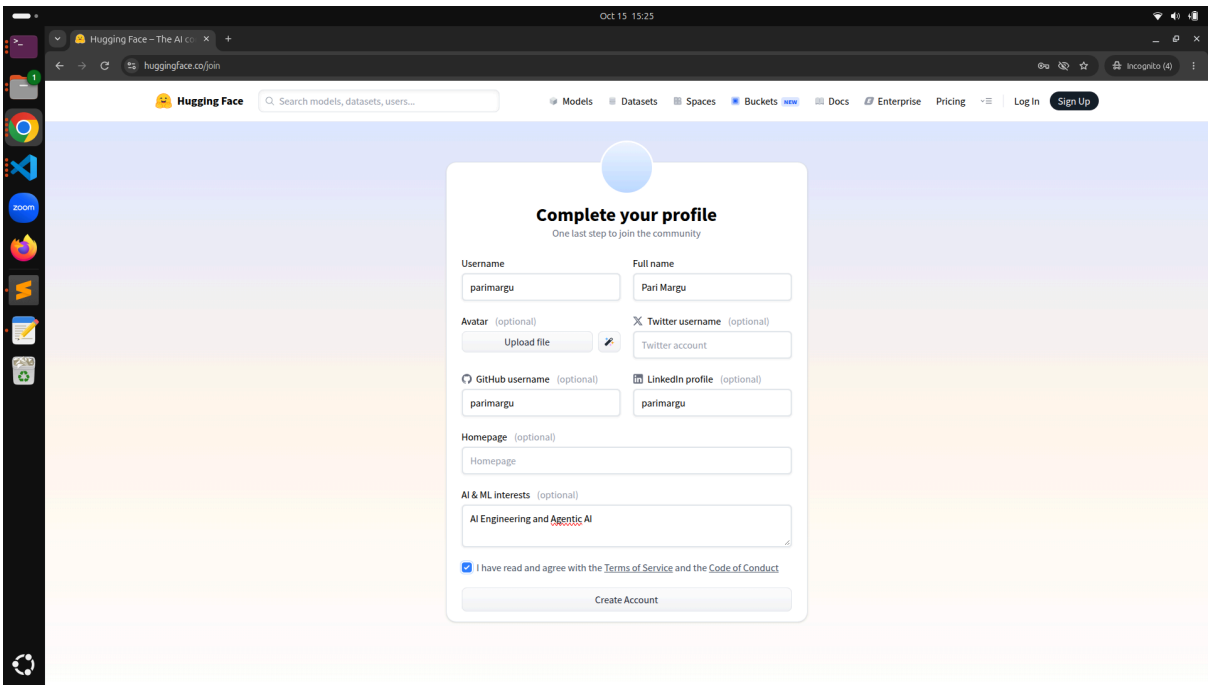
Hugging Face is a collaborative platform and open-source community often described as the "GitHub for machine learning".

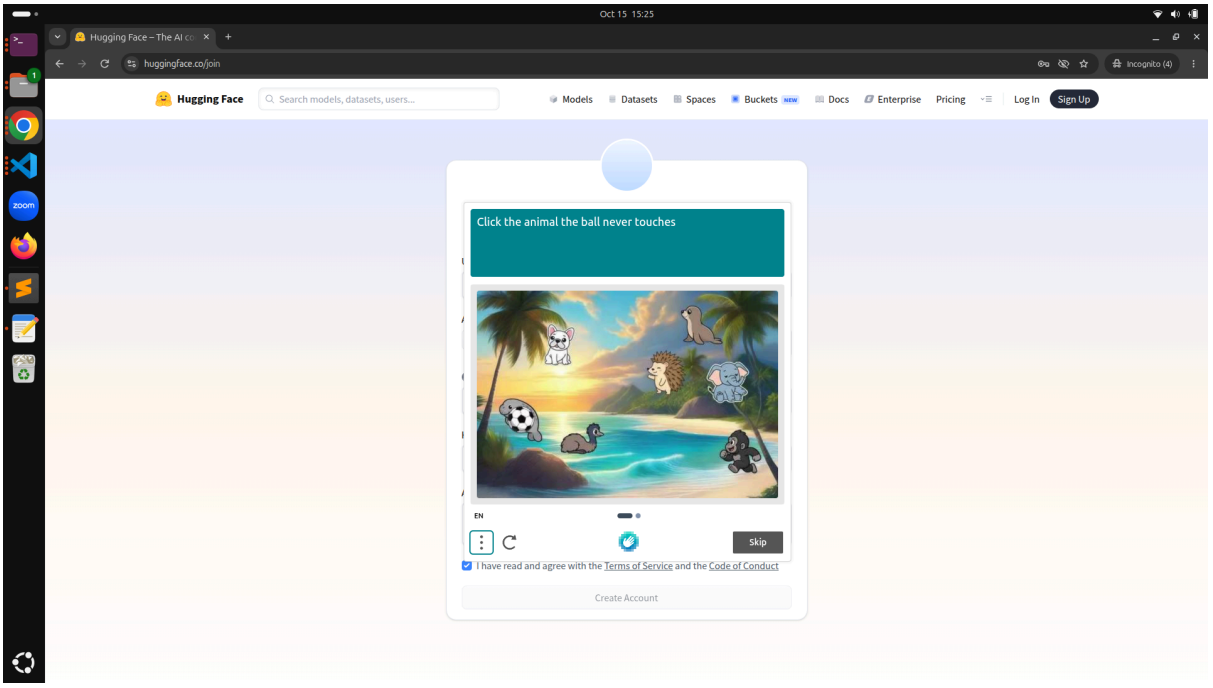
Url: [https://huggingface.co/](https://huggingface.co/)

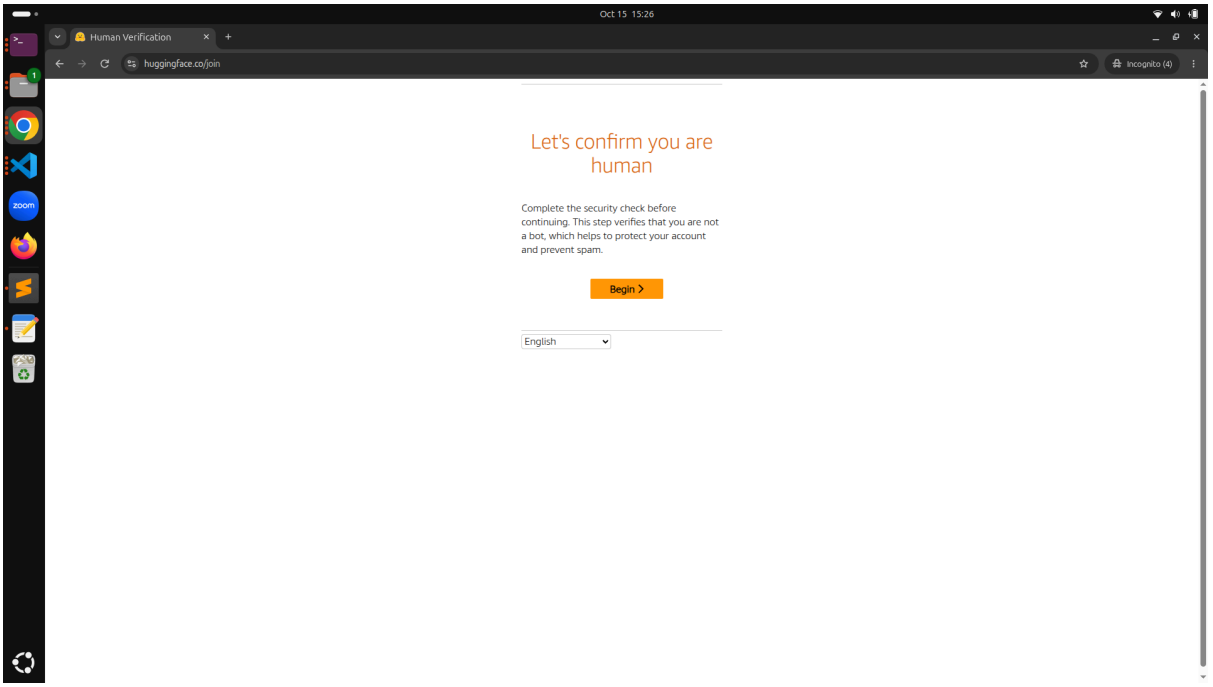


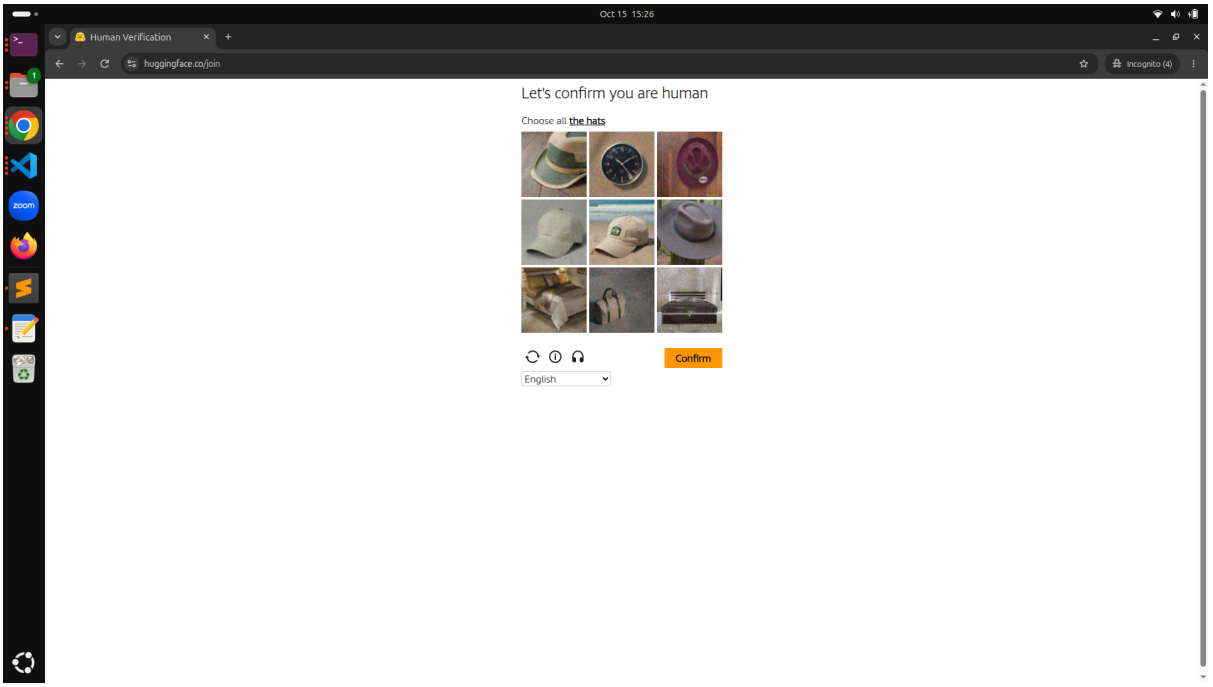

Oct 15 15:17
Hugging Face – The AI co
huggingface.co/join
Incognito (4)
Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets NEW
Docs
Enterprise
Pricing
Log In
Sign Up
Join Hugging Face
Join the community of machine learners!
Email Address
Email Address
Hint: Use your organization email to easily find and join your company/team org.
Password
Password
Next
Already have an account? Log in
Unlock SSO with Team & Enterprise

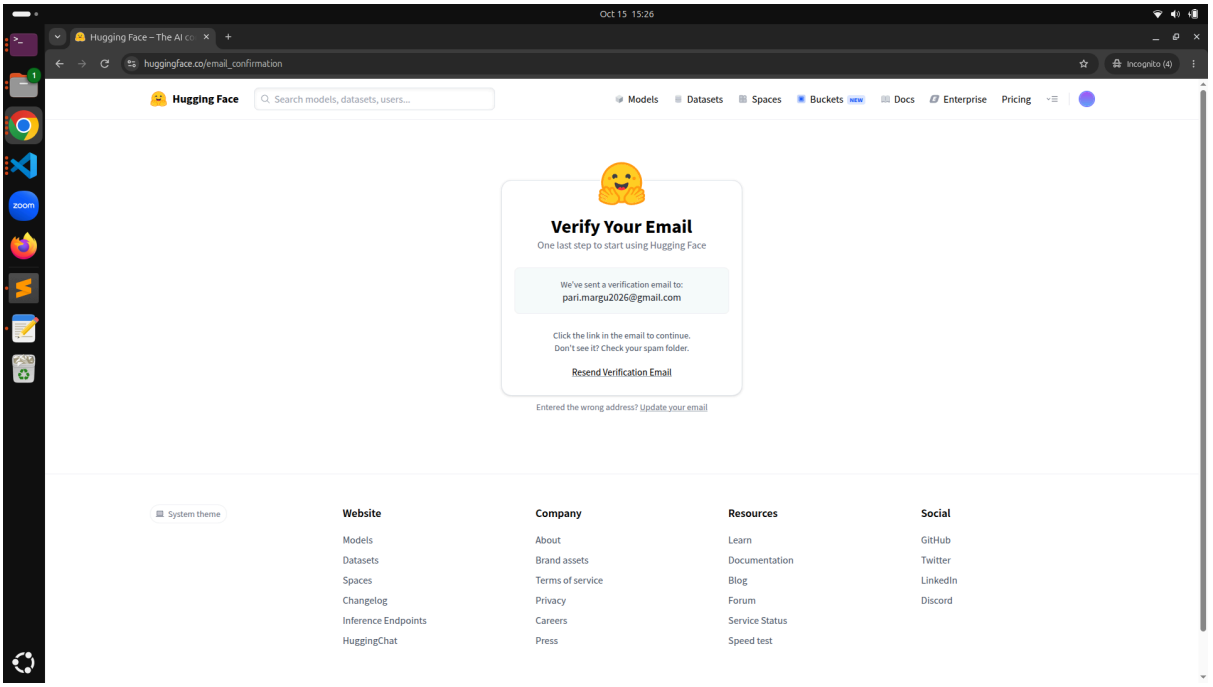

Oct 15 15:23
Hugging Face – The AI co
huggingface.co/join
Incognito (4)
Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets NEW
Docs
Enterprise
Pricing
Log In
Sign Up
Join Hugging Face
Join the community of machine learners!
Email Address
pari.margu2026@gmail.com
Hint: Use your organization email to easily find and join your company/team org.
Password
Must contain at least 8 characters
Must contain uppercase, lowercase letters, and numbers
If less than 12 characters, must contain a special character
Next
Already have an account? Log in
Unlock SSO with Team & Enterprise

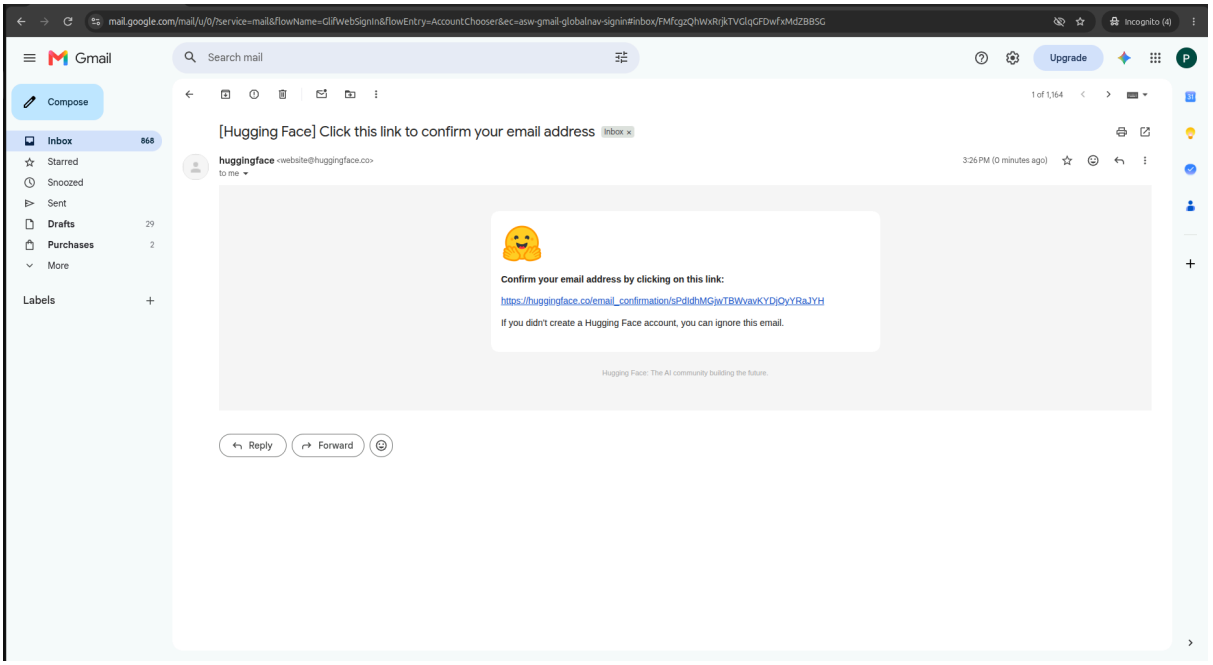

Oct 15 15:25
Hugging Face – The AI co
huggingface.co/join
Incognito (4)
Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets
Docs
Enterprise
Pricing
Log In
Sign Up
Complete your profile
One last step to join the community
Username
parimargu
Full name
Pari Margu
Avatar (optional)
Upload file
Twitter username (optional)
Twitter account
GitHub username (optional)
parimargu
LinkedIn profile (optional)
parimargu
Homepage (optional)
Homepage
AI & ML interests (optional)
AI Engineering and Agentic AI
I have read and agree with the Terms of Service and the Code of Conduct
Create Account

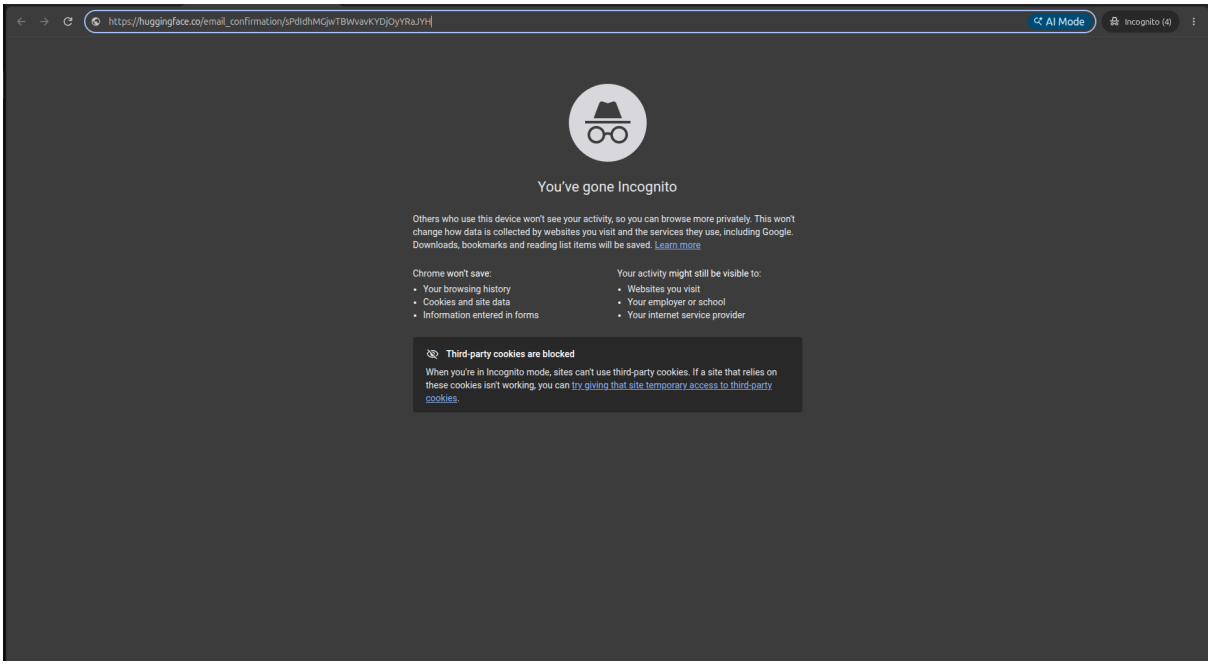

Oct 15 15:25
Hugging Face – The AI co
huggingface.co/join
Incognito (4)
Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets
Docs
Enterprise
Pricing
Log In
Sign Up
Click the animal the ball never touches
EN
Skip
I have read and agree with the Terms of Service and the Code of Conduct
Create Account

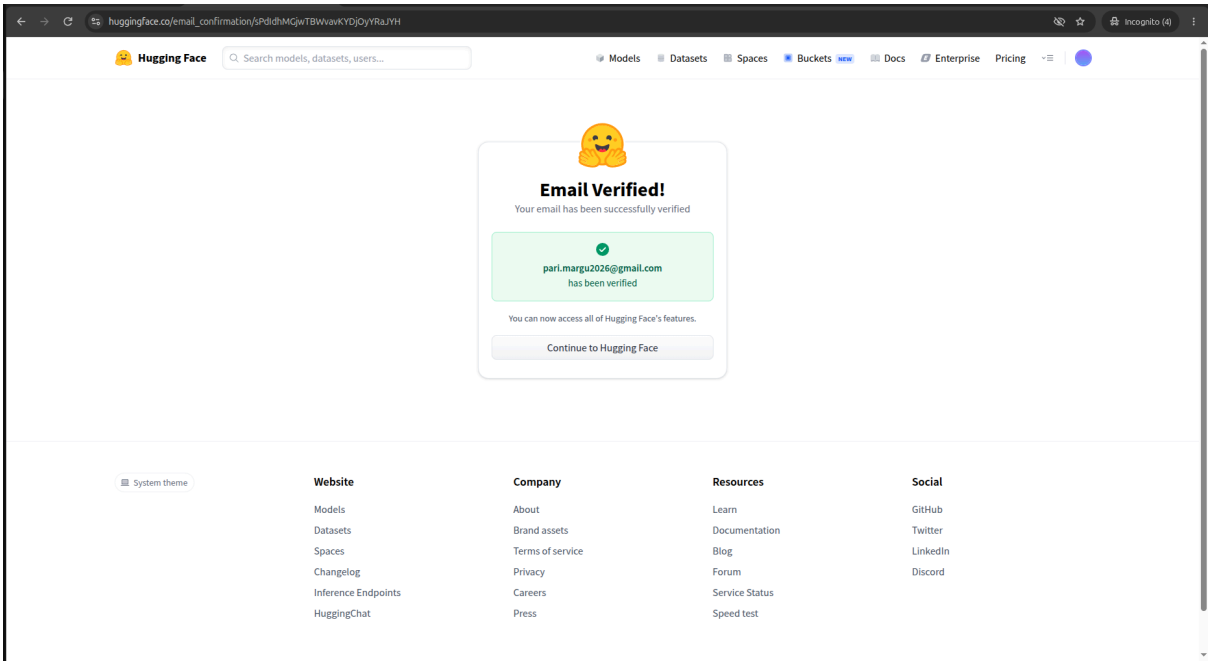

Oct 15 15:26
Human Verification
huggingface.co/join
Incognito (4)
Let's confirm you are human
Complete the security check before continuing. This step verifies that you are not a bot, which helps to protect your account and prevent spam.
Begin
English

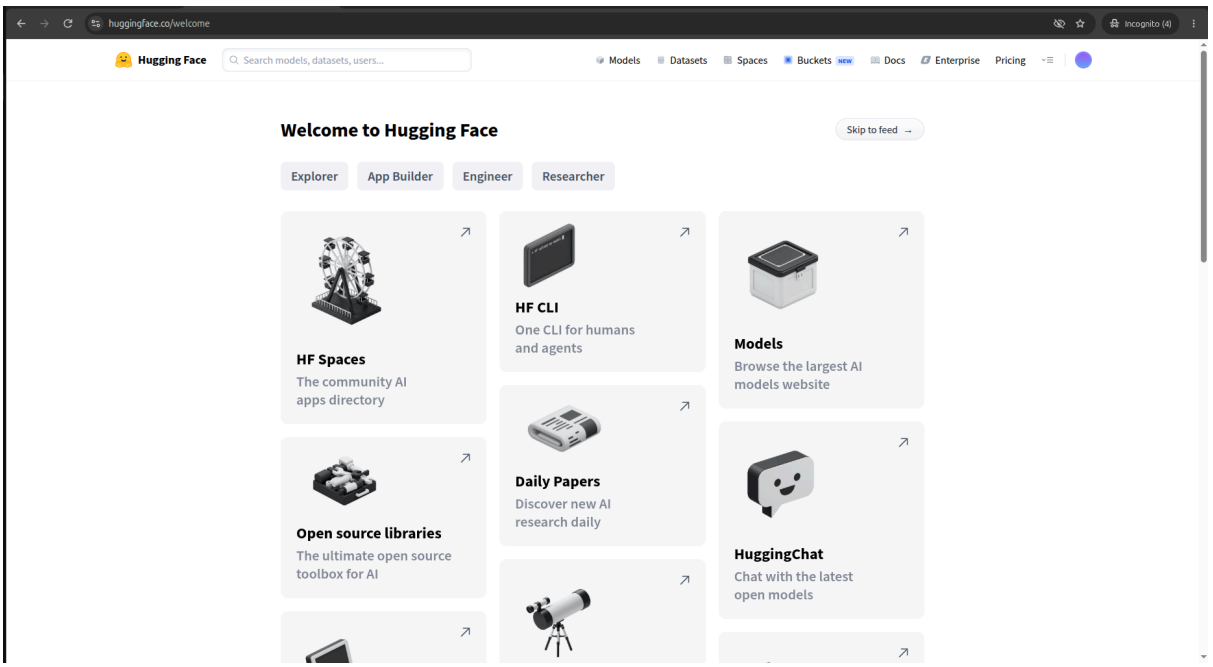

Oct 15 15:26
Human Verification
huggingface.co/join
Incognito (4)
Let's confirm you are human
Choose all the hats
Confirm
English

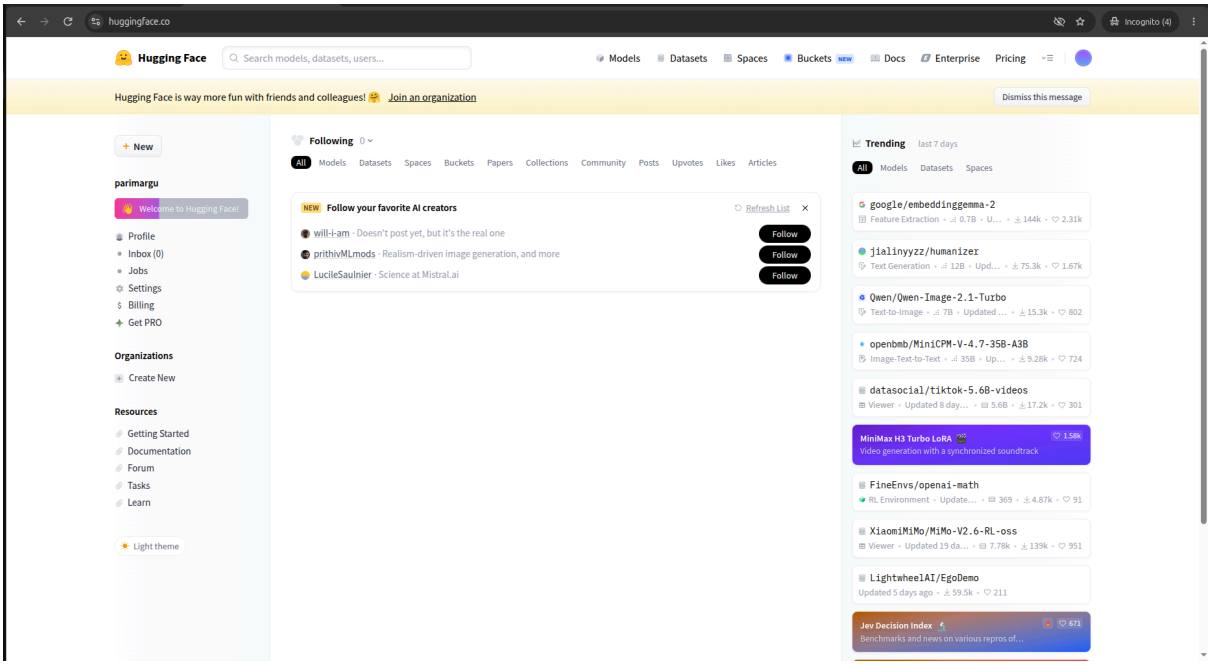

Oct 15 15:26
Hugging Face – The AI co
huggingface.co/email_confirmation
Incognito (4)
Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets NEW
Docs
Enterprise
Pricing
Verify Your Email
One last step to start using Hugging Face
We've sent a verification email to:
pari.margu2026@gmail.com
Click the link in the email to continue.
Don't see it? Check your spam folder.
Resend Verification Email
Entered the wrong address? Update your email
System theme
Website
Models
Datasets
Spaces
Changelog
Inference Endpoints
HuggingChat
Company
About
Brand assets
Terms of service
Privacy
Careers
Press
Resources
Learn
Documentation
Blog
Forum
Service Status
Speed test
Social
GitHub
Twitter
LinkedIn
Discord

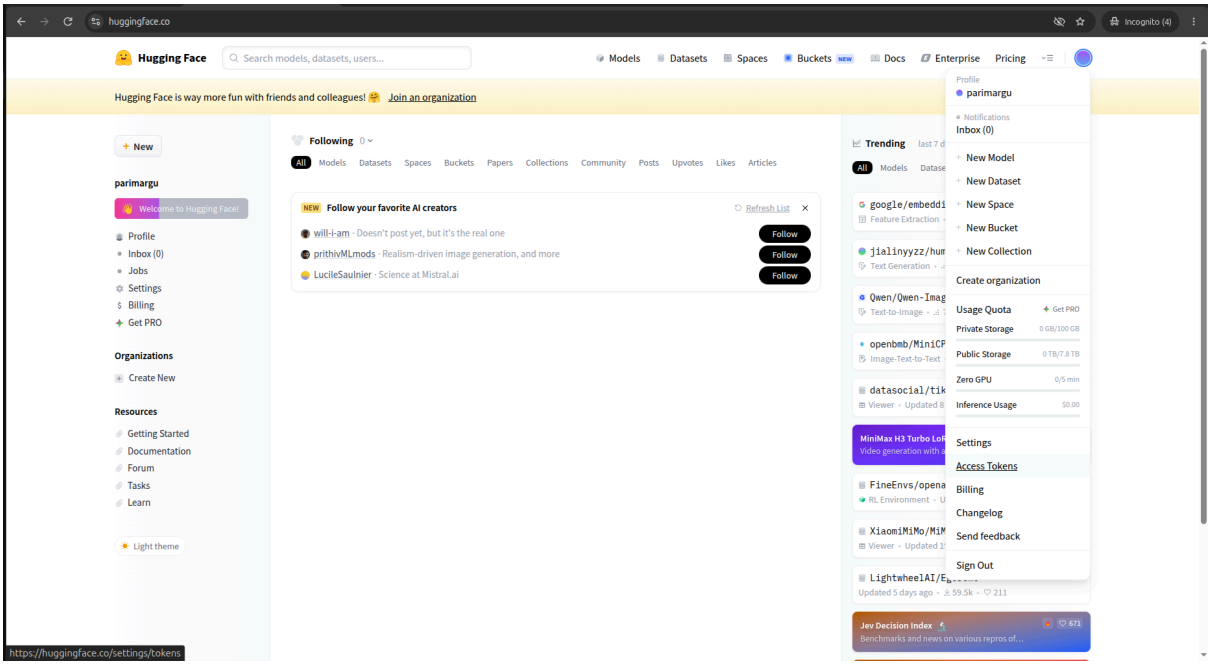

mail.google.com/mail/u/0/?service=mail&flowName=GlifWebSignIn&flowEntry=AccountChooser&ec=asw-gmail-globalnav-signin#inbox/FMfcgzQhlWxRrjkTVClqGFDwfxMdZBBSG
Incognito (4)
Gmail
Search mail
Upgrade
Compose
Inbox 868
Starred
Snoozed
Sent
Drafts 29
Purchases 2
More
Labels
1 of 1,164
[Hugging Face] Click this link to confirm your email address
Inbox ×
huggingface <website@huggingface.co>
to me
3:26 PM (0 minutes ago)
Confirm your email address by clicking on this link:
https://huggingface.co/email_confirmation/sPdIdhMGjwTBWvavKYDjOyYRaJYH
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Hugging Face: The AI community building the future.
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Forward

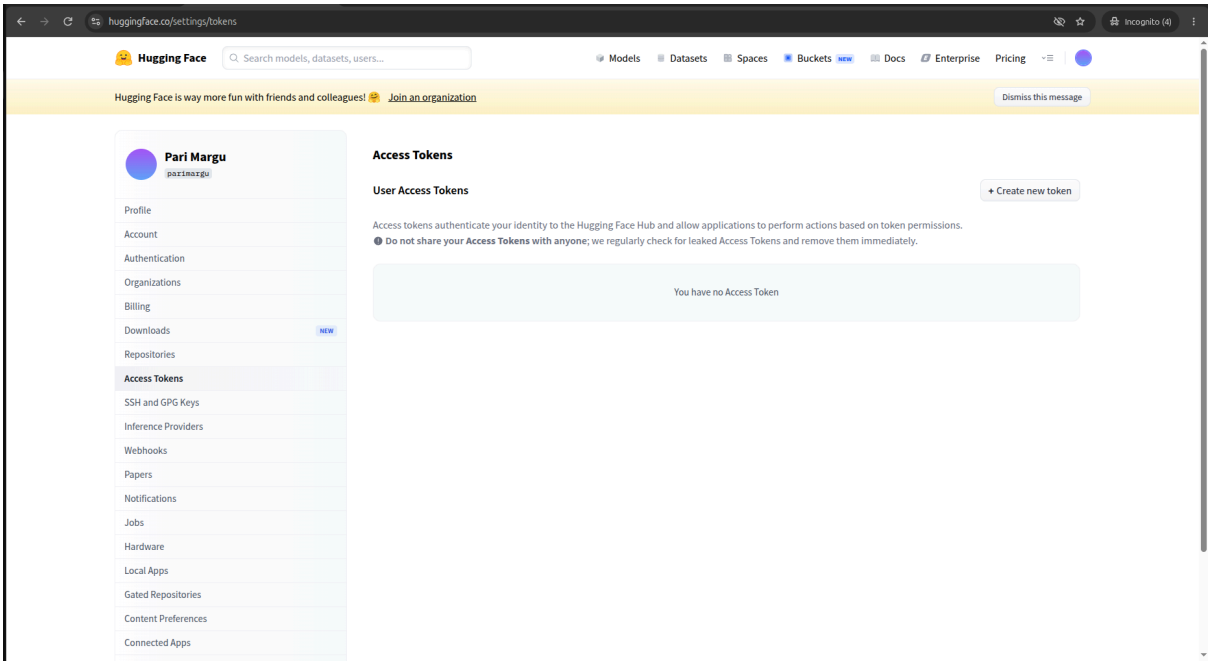

https://huggingface.co/email_confirmation/sPdIdhMGjwTBWvavKYDjOyYRaJYH
AI Mode
Incognito (4)
You've gone Incognito
Others who use this device won't see your activity, so you can browse more privately. This won't change how data is collected by websites you visit and the services they use, including Google. Downloads, bookmarks and reading list items will be saved. Learn more
Chrome won't save:
Your browsing history
Cookies and site data
Information entered in forms
Your activity might still be visible to:
Websites you visit
Your employer or school
Your internet service provider
Third-party cookies are blocked
When you're in Incognito mode, sites can't use third-party cookies. If a site that relies on these cookies isn't working, you can try giving that site temporary access to third-party cookies.

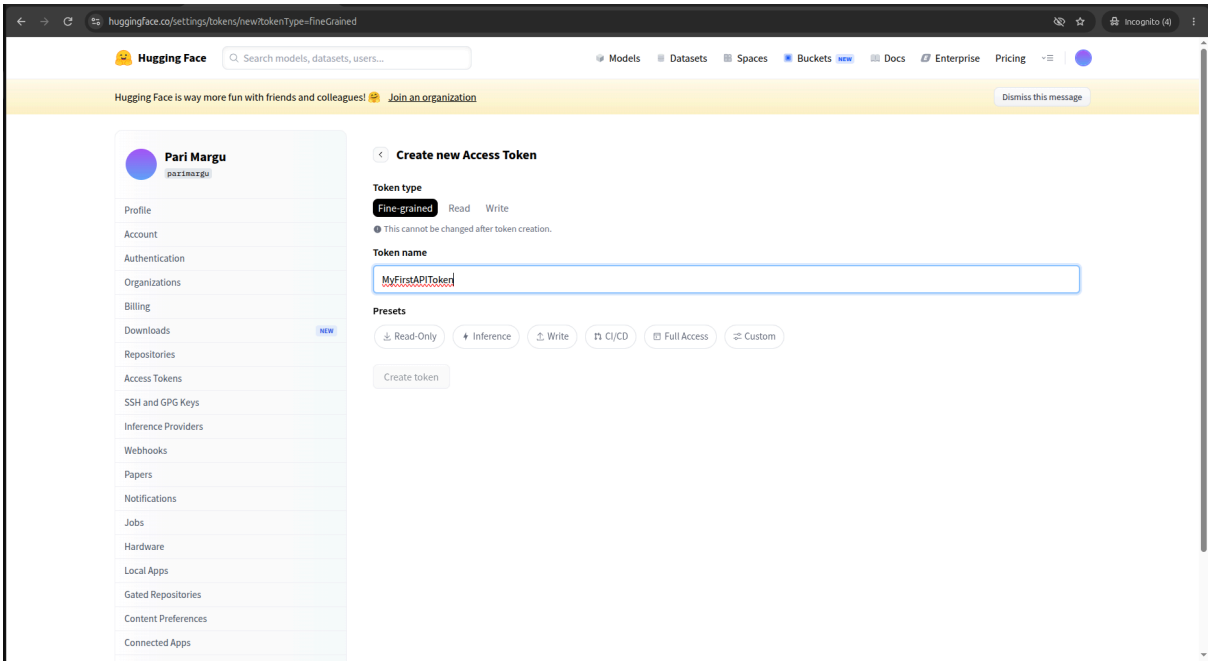

huggingface.co/email_confirmation/sPdIdhMGjwTBWvavKYDjOyYRaJYH
Incognito (4)
Hugging Face
Search models, datasets, users...
Models
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Spaces
Buckets NEW
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Your email has been successfully verified
pari.margu2026@gmail.com
has been verified
You can now access all of Hugging Face's features.
Continue to Hugging Face
System theme
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About
Brand assets
Terms of service
Privacy
Careers
Press
Resources
Learn
Documentation
Blog
Forum
Service Status
Speed test
Social
GitHub
Twitter
LinkedIn
Discord

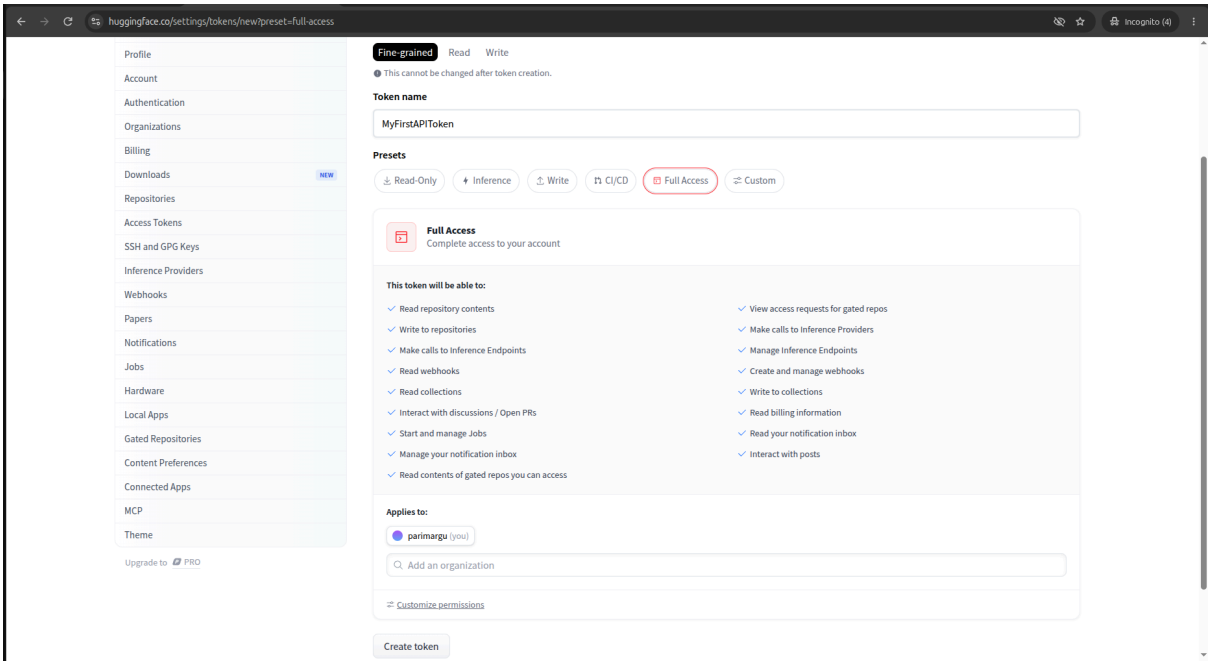

huggingface.co/welcome
Incognito (4)
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Docs
Enterprise
Pricing
Welcome to Hugging Face
Skip to feed
Explorer
App Builder
Engineer
Researcher
HF Spaces
The community AI apps directory
HF CLI
One CLI for humans and agents
Models
Browse the largest AI models website
Open source libraries
The ultimate open source toolbox for AI
Daily Papers
Discover new AI research daily
HuggingChat
Chat with the latest open models

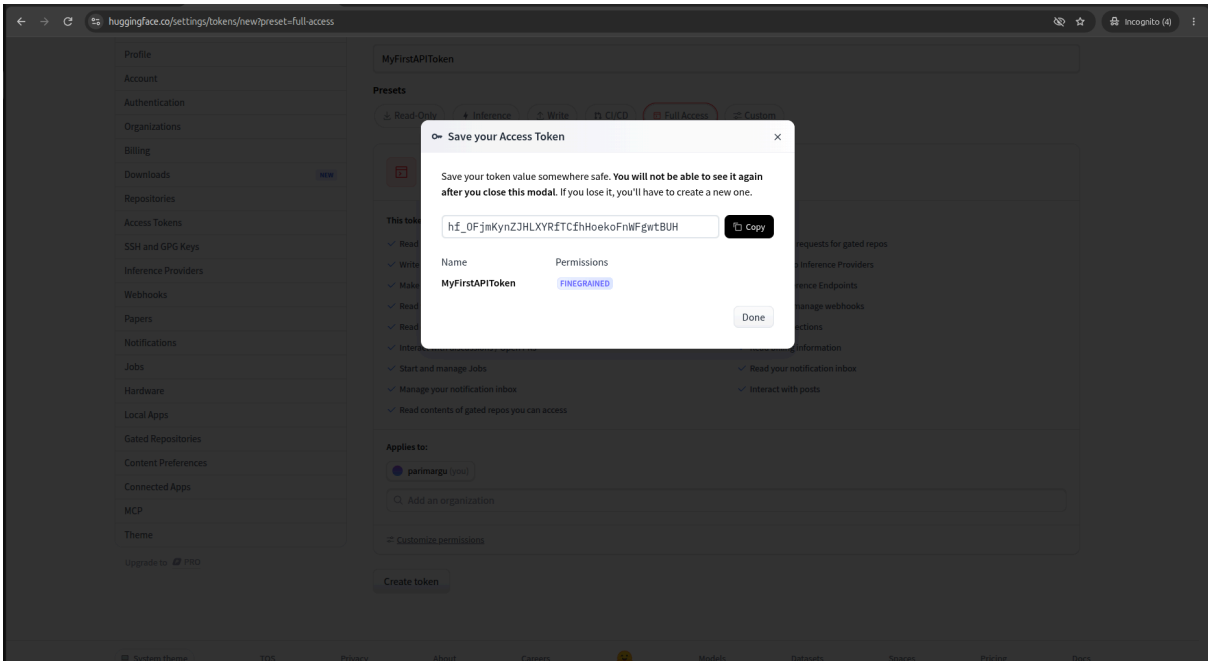

huggingface.co
Incognito (4)
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Documentation
Forum
Tasks
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Following 0
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Models
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Spaces
Buckets
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Posts
Upvotes
Likes
Articles
NEW
Follow your favorite AI creators
Refresh List
will-i-am · Doesn't post yet, but it's the real one
prithivMLmods · Realism-driven image generation, and more
LucileSaulnier · Science at Mistral.ai
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jialinyyzz/humanizer
Qwen/Qwen-Image-2.1-Turbo
openbmb/MiniCPM-V-4.7-35B-A3B
datasocial/tiktok-5.6B-videos
MiniMax H3 Turbo LoRA
Video generation with a synchronized soundtrack
FineEnvs/openai-math
XiaomiMiMo/MiMo-V2.6-RL-oss
LightwheelAI/EgoDemo
Jev Decision Index
Benchmarks and news on various repros of...

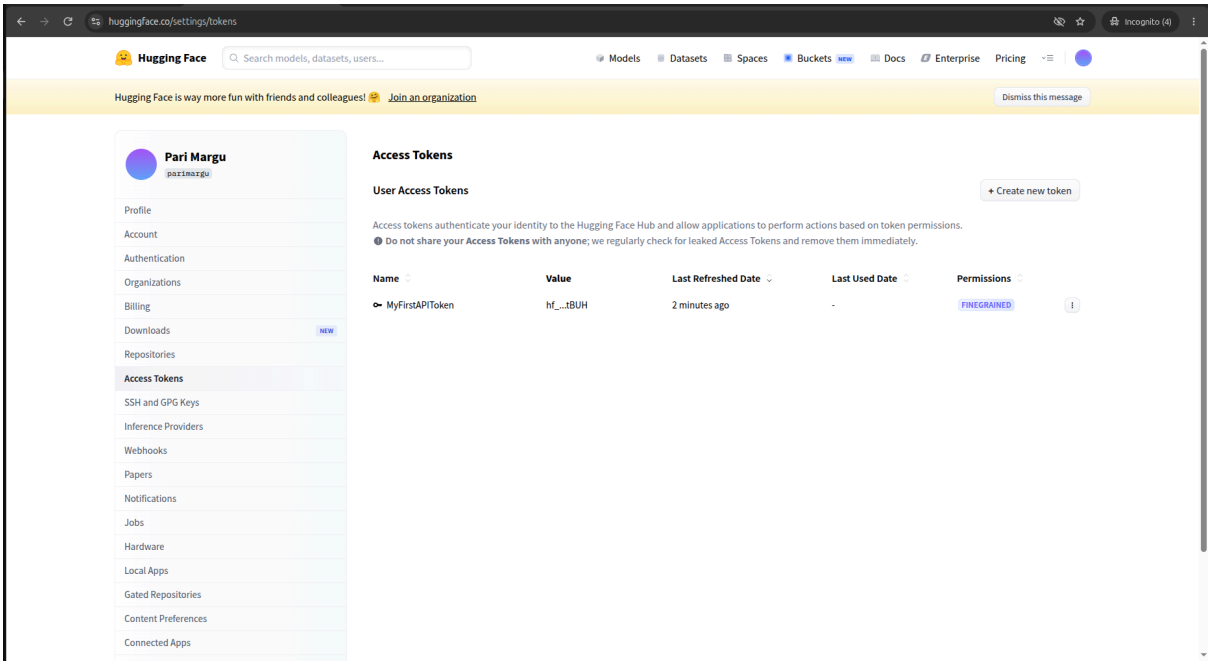

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Documentation
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All
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Buckets
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Likes
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will-i-am · Doesn't post yet, but it's the real one
prithivMLmods · Realism-driven image generation, and more
LucileSaulnier · Science at Mistral.ai
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Inbox (0)
New Model
New Dataset
New Space
New Bucket
New Collection
Create organization
Usage Quota
Get PRO
Private Storage
0 GB/100 GB
Public Storage
0 TB/7.8 TB
Zero GPU
0/5 min
Inference Usage
$0.00
Settings
Access Tokens
Billing
Changelog
Send feedback
Sign Out
Jev Decision Index
Benchmarks and news on various repros of...
https://huggingface.co/settings/tokens

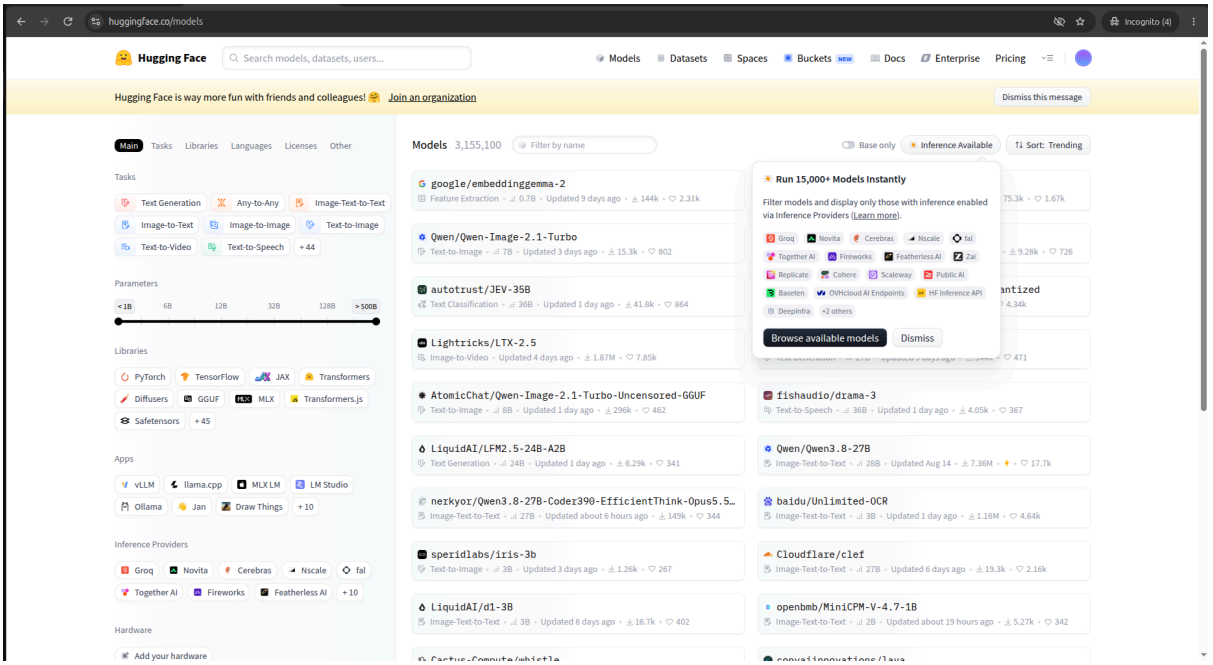

huggingface.co/settings/tokens
Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets
Docs
Enterprise
Pricing
Incognito (4)
Hugging Face is way more fun with friends and colleagues! 🤗 Join an organization
Dismiss this message
Pari Margu
parimargu
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Authentication
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Billing
Downloads
NEW
Repositories
Access Tokens
SSH and GPG Keys
Inference Providers
Webhooks
Papers
Notifications
Jobs
Hardware
Local Apps
Gated Repositories
Content Preferences
Connected Apps
Access Tokens
User Access Tokens
+ Create new token
Access tokens authenticate your identity to the Hugging Face Hub and allow applications to perform actions based on token permissions.
Do not share your Access Tokens with anyone; we regularly check for leaked Access Tokens and remove them immediately.
You have no Access Token


huggingface.co/settings/tokens/new?tokenType=fineGrained
Incognito (4)
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Spaces
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Docs
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Pricing
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Downloads NEW
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Access Tokens
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Webhooks
Papers
Notifications
Jobs
Hardware
Local Apps
Gated Repositories
Content Preferences
Connected Apps
Create new Access Token
Token type
Fine-grained
Read
Write
This cannot be changed after token creation.
Token name
MyFirstAPIToken
Presets
Read-Only
Inference
Write
CI/CD
Full Access
Custom
Create token

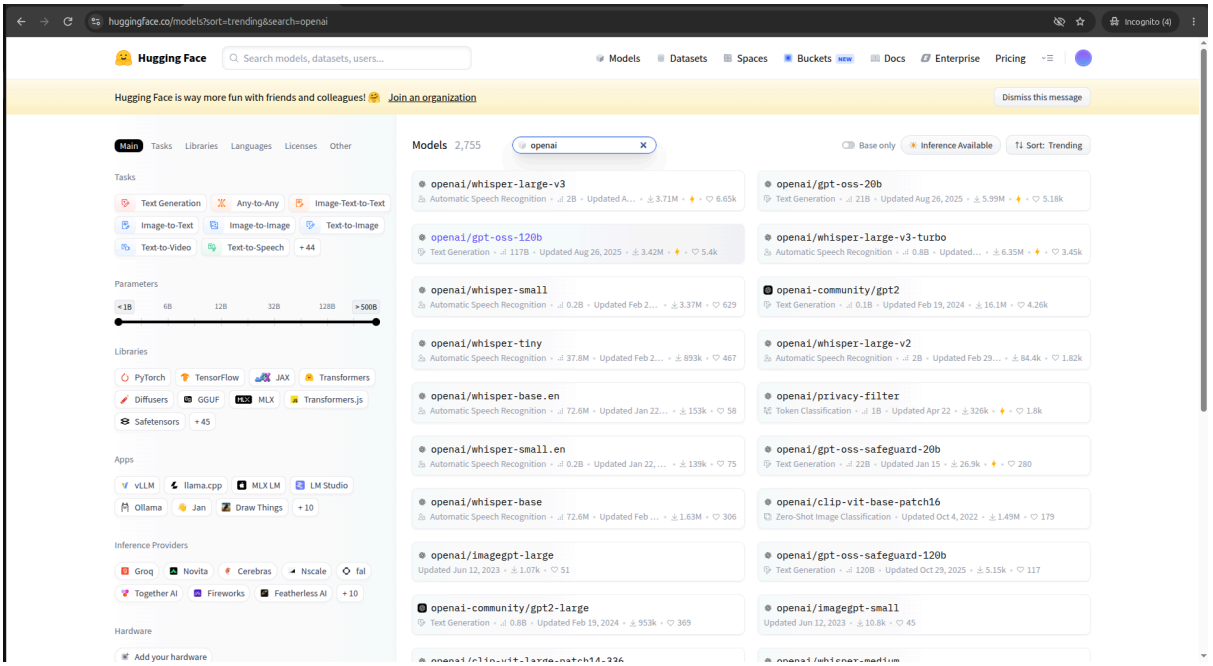

huggingface.co/settings/tokens/new?preset=full-access
Incognito (4)
Profile
Account
Authentication
Organizations
Billing
Downloads NEW
Repositories
Access Tokens
SSH and GPG Keys
Inference Providers
Webhooks
Papers
Notifications
Jobs
Hardware
Local Apps
Gated Repositories
Content Preferences
Connected Apps
MCP
Theme
Upgrade to PRO
Fine-grained
Read
Write
This cannot be changed after token creation.
Token name
MyFirstAPIToken
Presets
Read-Only
Inference
Write
CI/CD
Full Access
Custom
Full Access
Complete access to your account
This token will be able to:
Read repository contents
Write to repositories
Make calls to Inference Endpoints
Read webhooks
Read collections
Interact with discussions / Open PRs
Start and manage Jobs
Manage your notification inbox
Read contents of gated repos you can access
View access requests for gated repos
Make calls to Inference Providers
Manage Inference Endpoints
Create and manage webhooks
Write to collections
Read billing information
Read your notification inbox
Interact with posts
Applies to:
parimargu (you)
Add an organization
Customize permissions
Create token

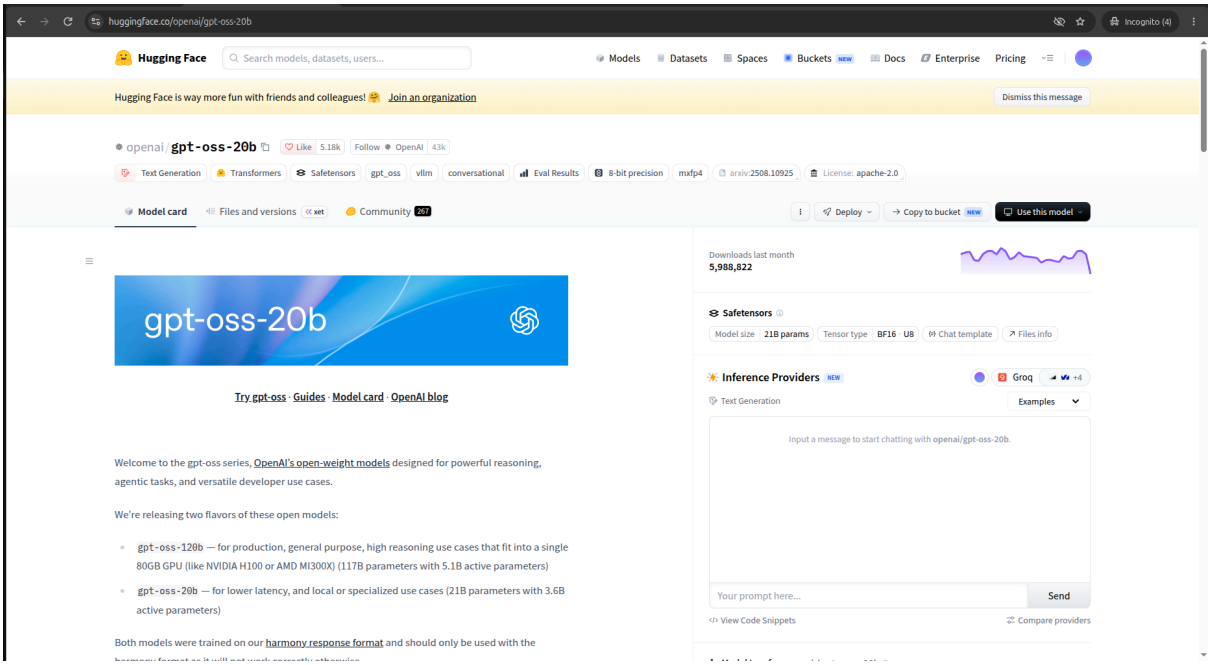

huggingface.co/settings/tokens/new?preset=full-access
Incognito (4)
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Local Apps
Gated Repositories
Content Preferences
Connected Apps
MCP
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MyFirstAPIToken
Presets
Save your Access Token
Save your token value somewhere safe. You will not be able to see it again after you close this modal. If you lose it, you'll have to create a new one.
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
Copy
Name
Permissions
MyFirstAPIToken
FINEGRAINED
Done
Start and manage Jobs
Manage your notification inbox
Read contents of gated repos you can access
Read your notification inbox
Interact with posts
Applies to:
parimargu (you)
Add an organization
Customize permissions
Create token

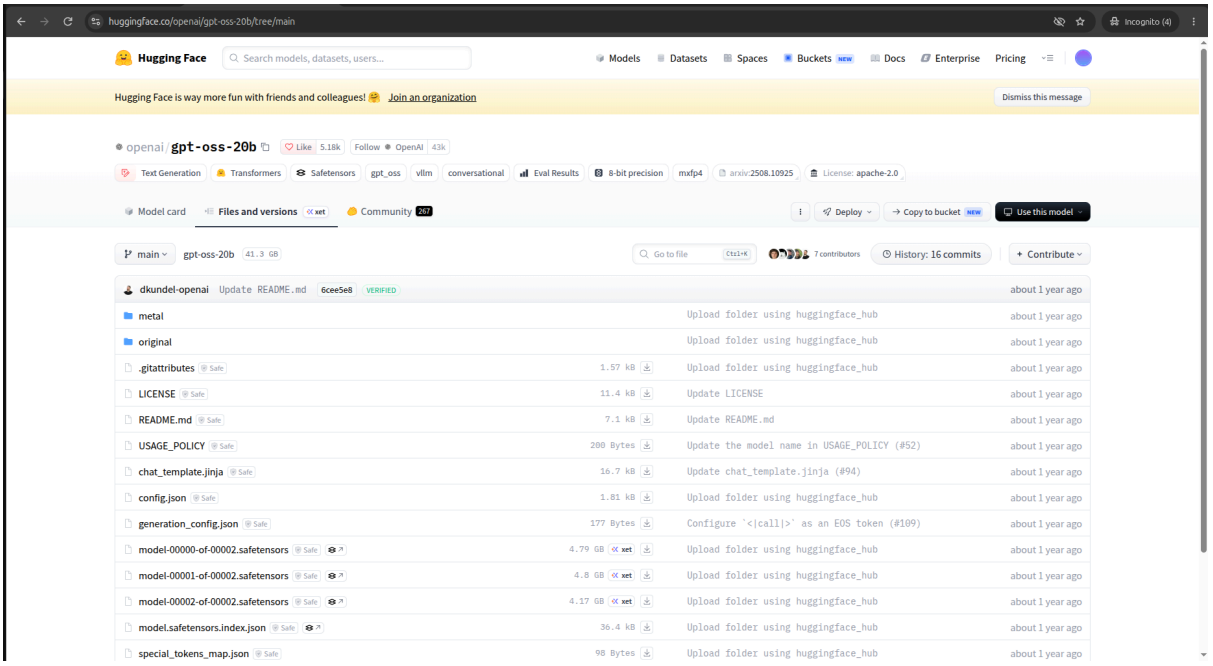

huggingface.co/settings/tokens
Incognito (4)
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Gated Repositories
Content Preferences
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Access Tokens
User Access Tokens
+ Create new token
Access tokens authenticate your identity to the Hugging Face Hub and allow applications to perform actions based on token permissions.
Do not share your Access Tokens with anyone; we regularly check for leaked Access Tokens and remove them immediately.
Name
Value
Last Refreshed Date
Last Used Date
Permissions
MyFirstAPIToken
hf_...tBUH
2 minutes ago
-
FINEGRAINED

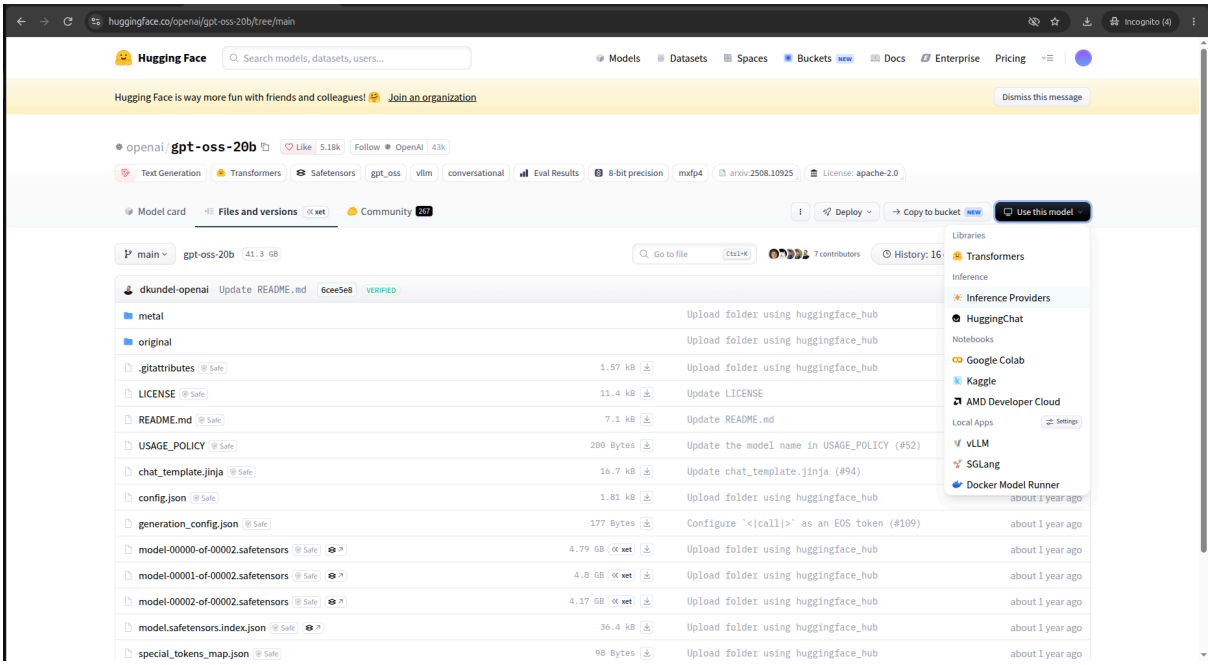


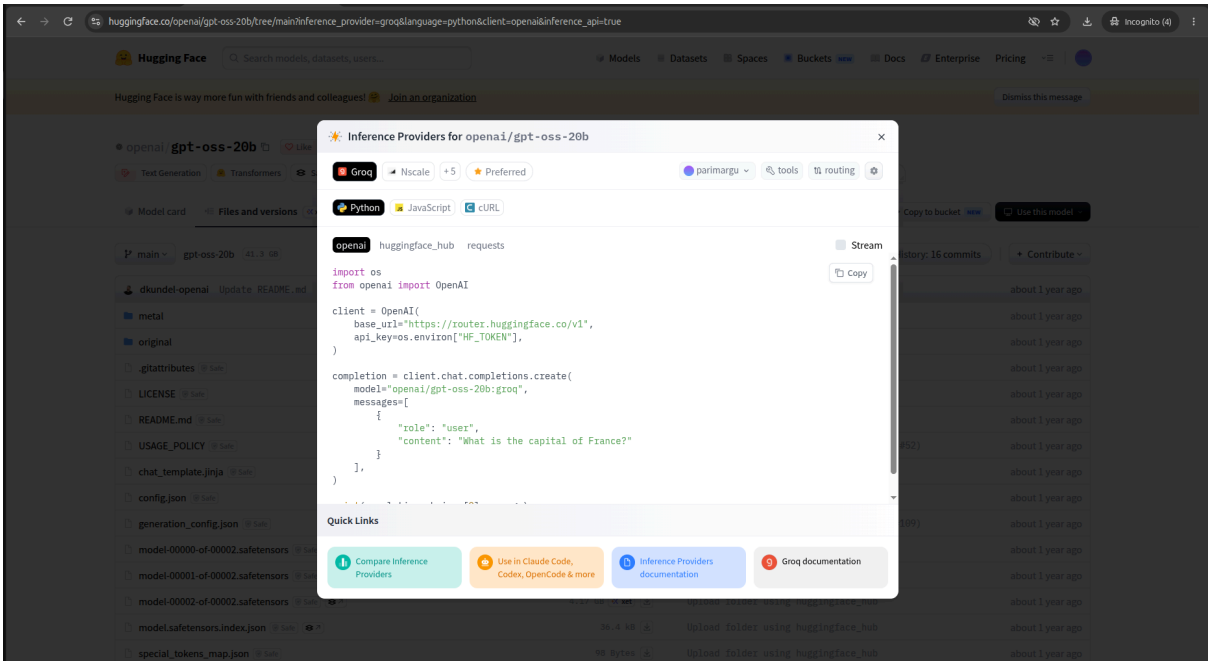

https://huggingface.co/models




huggingface.co/openai/gpt-oss-20b
Hugging Face
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Models
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Transformers
Safetensors
gpt_oss
vllm
conversational
Eval Results
8-bit precision
mxfp4
arxiv:2508.10925
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Model card
Files and versions
Community 267
Deploy
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Try gpt-oss · Guides · Model card · OpenAI blog
Welcome to the gpt-oss series, OpenAI's open-weight models designed for powerful reasoning, agentic tasks, and versatile developer use cases.
We're releasing two flavors of these open models:
gpt-oss-120b — for production, general purpose, high reasoning use cases that fit into a single 80GB GPU (like NVIDIA H100 or AMD MI300X) (117B parameters with 5.1B active parameters)
gpt-oss-20b — for lower latency, and local or specialized use cases (21B parameters with 3.6B active parameters)
Both models were trained on our harmony response format and should only be used with the
Downloads last month
5,988,822
Safetensors
Model size 21B params
Tensor type BF16 · U8
Chat template
Files info
Inference Providers
Groq
Text Generation
Examples
Input a message to start chatting with openai/gpt-oss-20b.
Your prompt here...
Send
View Code Snippets
Compare providers


huggingface.co/openai/gpt-oss-20b/tree/main
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vllm
conversational
Eval Results
8-bit precision
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Model card
Files and versions
Community 267
Deploy
Copy to bucket
Use this model
main
gpt-oss-20b 41.3 GB
Go to file
Ctrl+K
7 contributors
History: 16 commits
Contribute
dkundel-openai Update README.md 6cee5e8 VERIFIED about 1 year ago
metal Upload folder using huggingface_hub about 1 year ago
original Upload folder using huggingface_hub about 1 year ago
.gitattributes 1.57 kB Upload folder using huggingface_hub about 1 year ago
LICENSE 11.4 kB Update LICENSE about 1 year ago
README.md 7.1 kB Update README.md about 1 year ago
USAGE_POLICY 200 Bytes Update the model name in USAGE_POLICY (#52) about 1 year ago
chat_template.jinja 16.7 kB Update chat_template.jinja (#94) about 1 year ago
config.json 1.81 kB Upload folder using huggingface_hub about 1 year ago
generation_config.json 177 Bytes Configure `<|call|>` as an EOS token (#109) about 1 year ago
model-00000-of-00002.safetensors 4.79 GB Upload folder using huggingface_hub about 1 year ago
model-00001-of-00002.safetensors 4.8 GB Upload folder using huggingface_hub about 1 year ago
model-00002-of-00002.safetensors 4.17 GB Upload folder using huggingface_hub about 1 year ago
model.safetensors.index.json 36.4 kB Upload folder using huggingface_hub about 1 year ago
special_tokens_map.json 98 Bytes Upload folder using huggingface_hub about 1 year ago




Code Snippet

```
import os
from openai import OpenAI

client = OpenAI(
    base_url="https://router.huggingface.co/v1",
    api_key=os.environ["HF_TOKEN"],
)
```

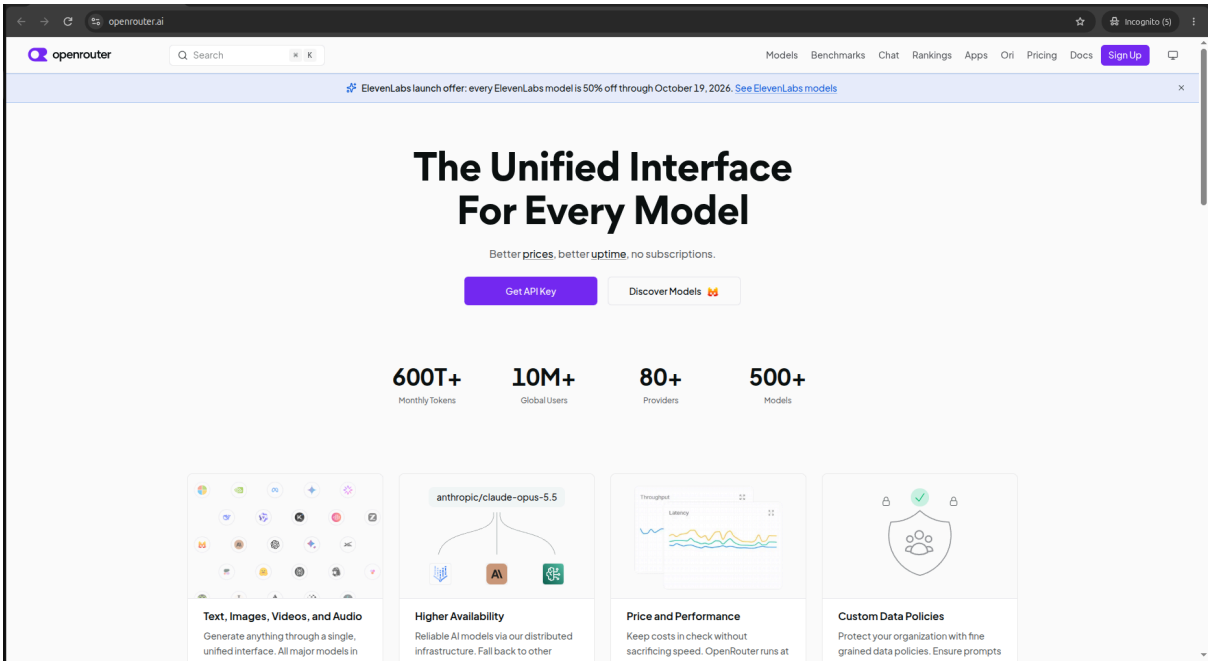
```
completion = client.chat.completions.create(
    model="openai/gpt-oss-20b:groq",
    messages=[
        {
            "role": "user",
            "content": "What is the capital of France?"
        }
    ],
)

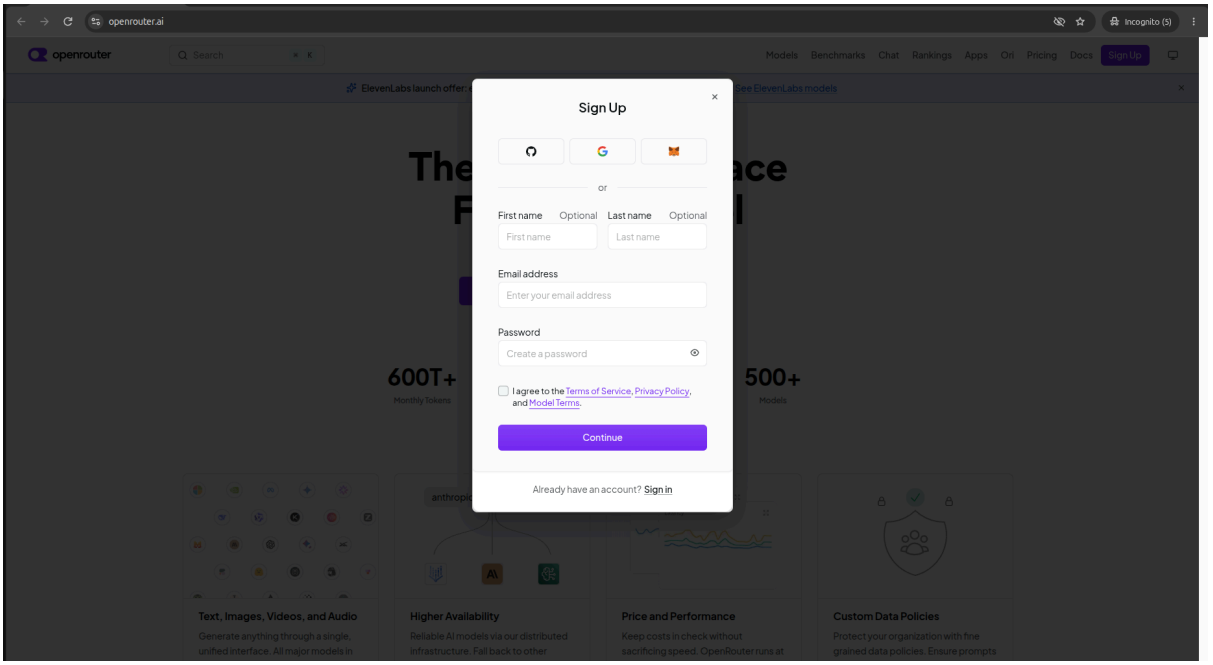
print(completion.choices[0].message)
```

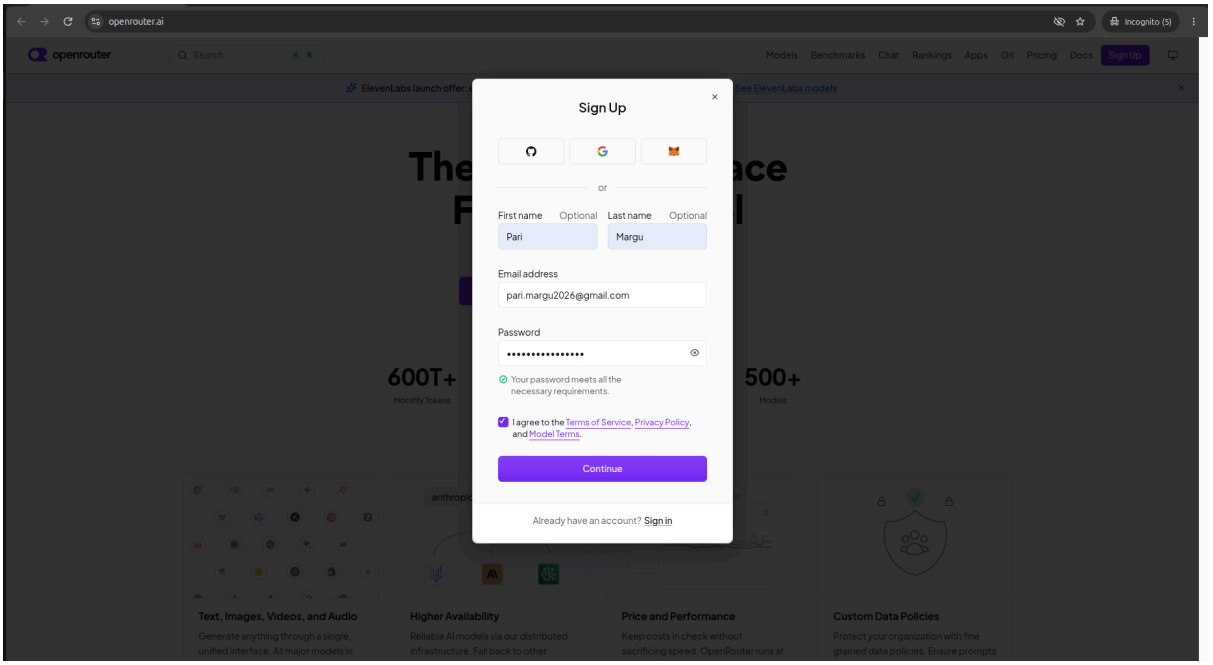
## How to connect OpenRouter’s LLM Model:

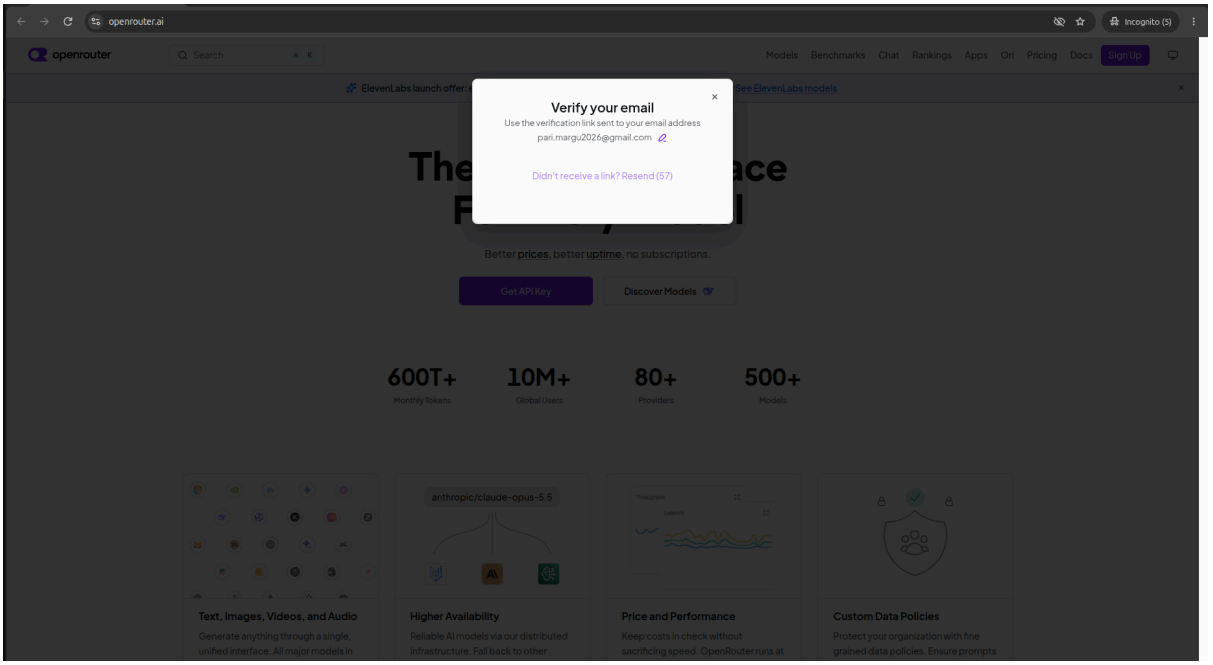
OpenRouter is a unified API gateway that lets you access hundreds of large language models (LLMs) from different providers using a single API key.

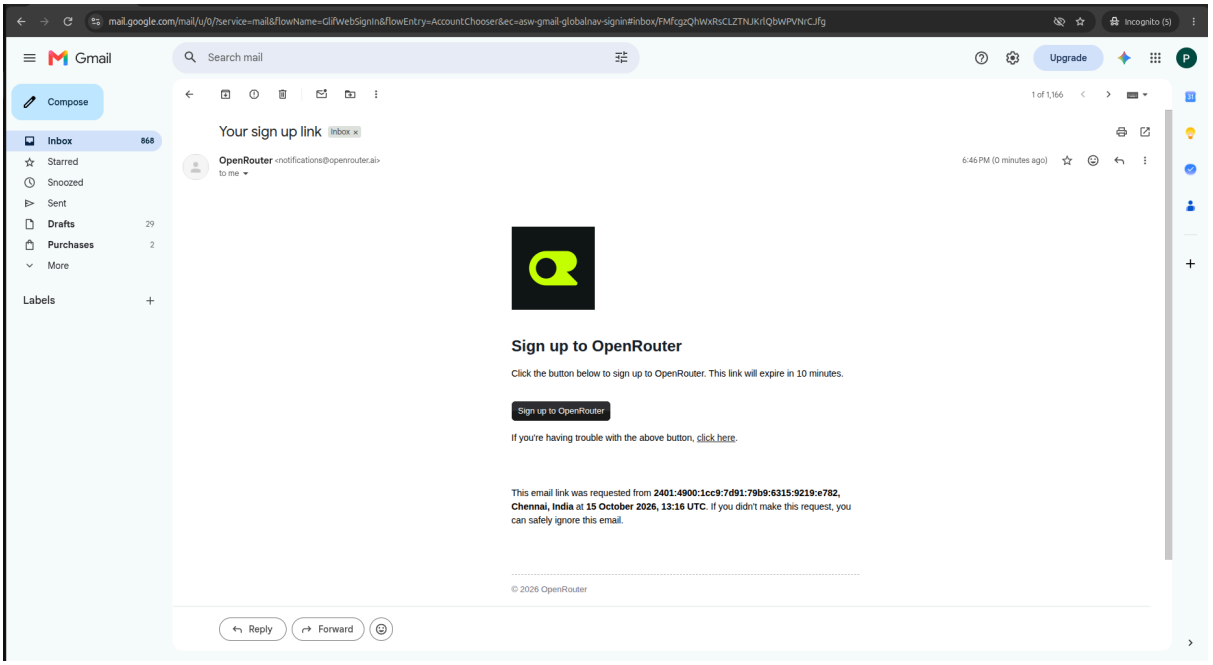
Url: [https://openrouter.ai/](https://openrouter.ai/)

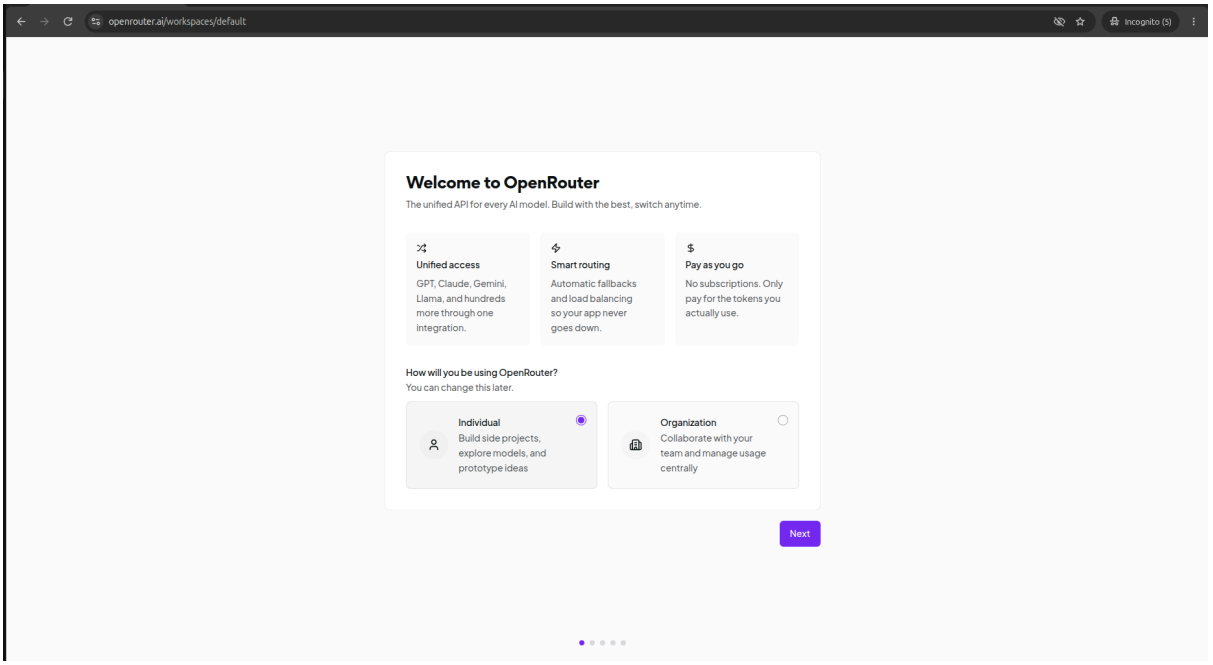


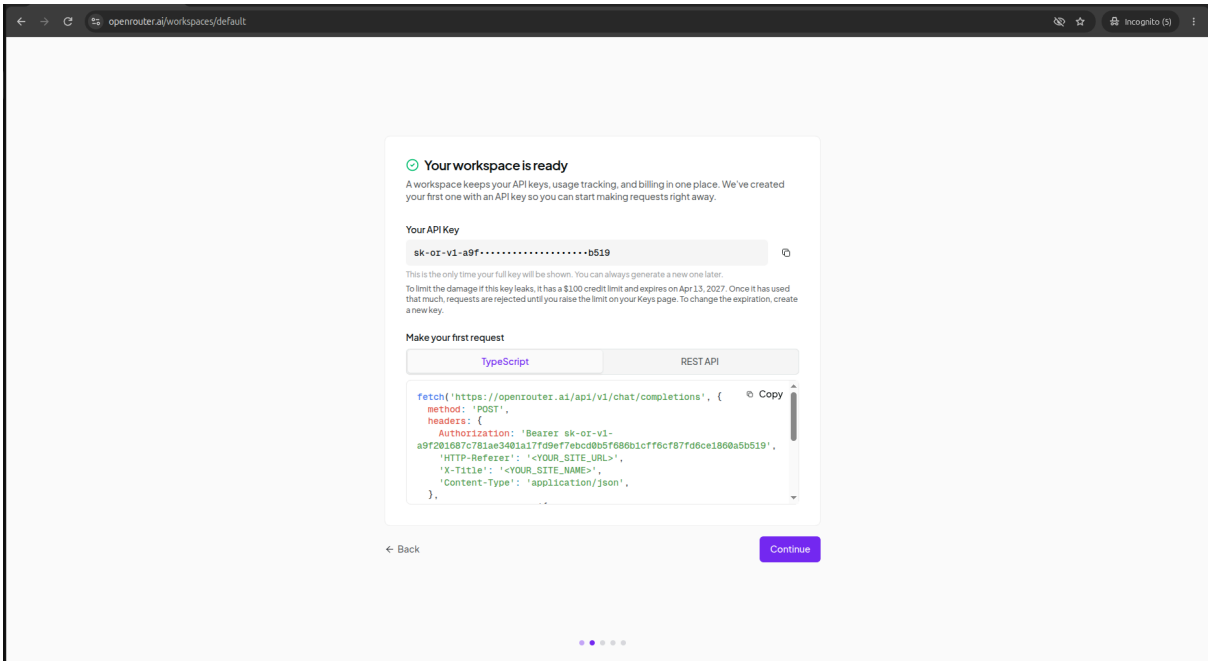

openrouter.ai
Incognito (5)
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Last name
Optional
Email address
Enter your email address
Password
Create a password
I agree to the Terms of Service, Privacy Policy, and Model Terms.
Continue
Already have an account? Sign in
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Monthly Tokens
500+
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Text, Images, Videos, and Audio
Higher Availability
Price and Performance
Custom Data Policies

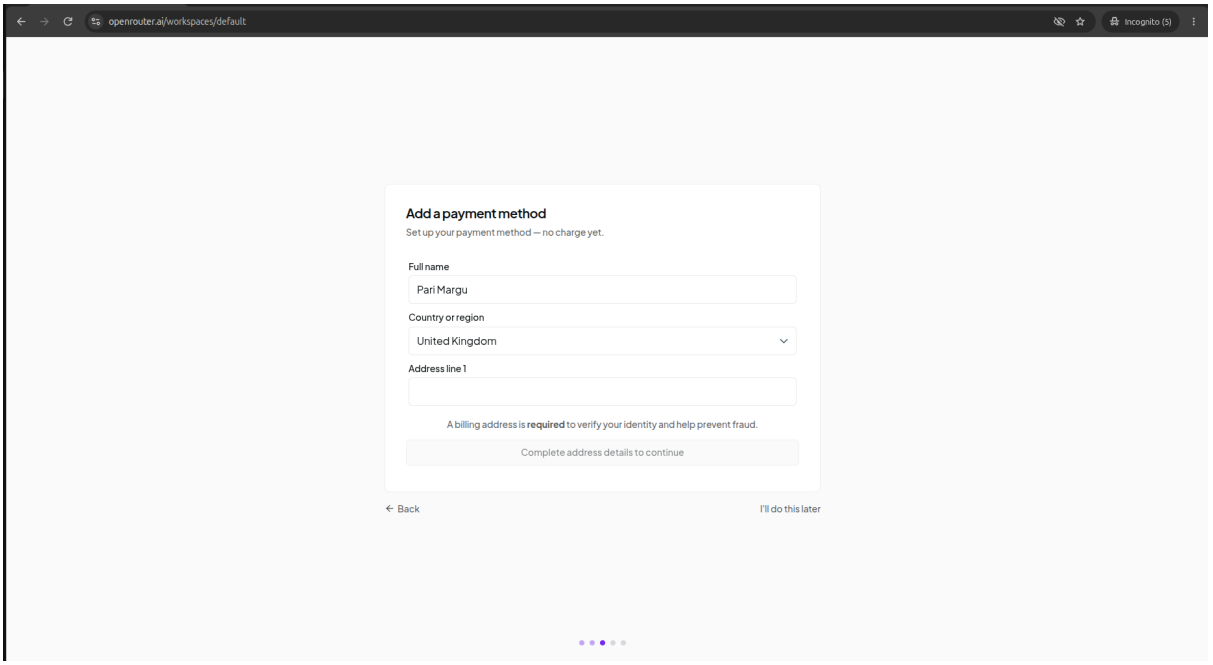

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pari.margu2026@gmail.com
Password
Your password meets all the necessary requirements.
I agree to the Terms of Service, Privacy Policy, and Model Terms.
Continue
Already have an account? Sign in
600T+
Monthly Tokens
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Models
Text, Images, Videos, and Audio
Higher Availability
Price and Performance
Custom Data Policies

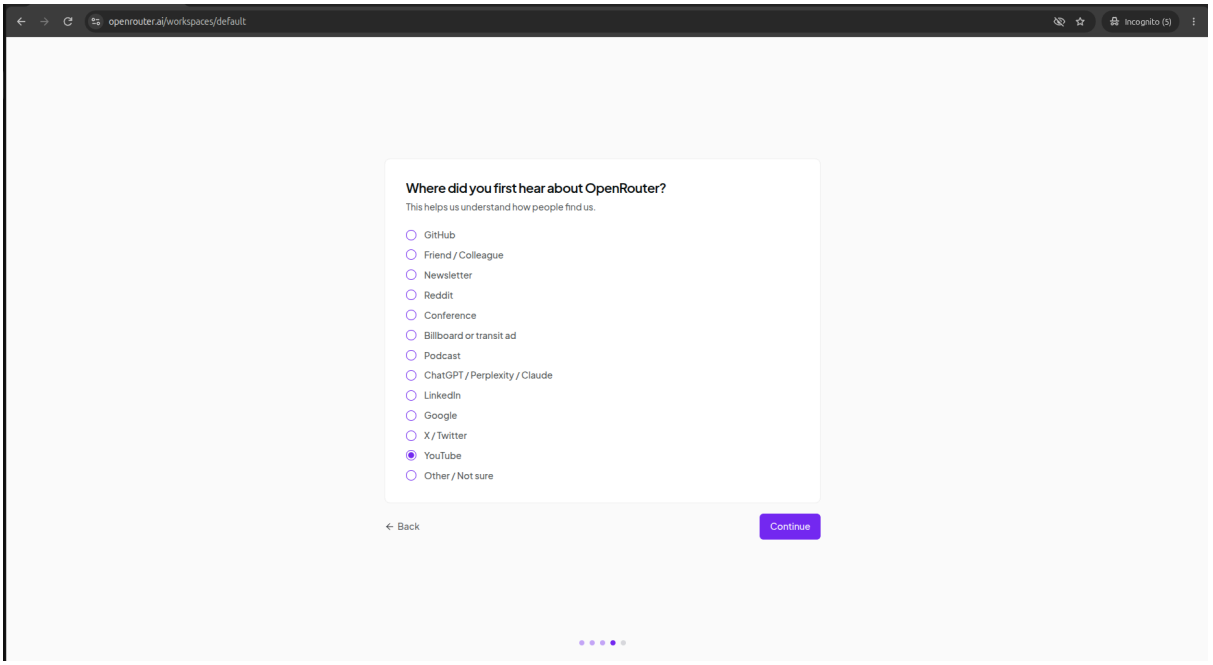

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Rankings
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Pricing
Docs
Sign Up
Verify your email
Use the verification link sent to your email address
pari.margu2026@gmail.com
Didn't receive a link? Resend (57)
Better prices, better uptime, no subscriptions.
Get API Key
Discover Models
600T+
Monthly Tokens
10M+
Global Users
80+
Providers
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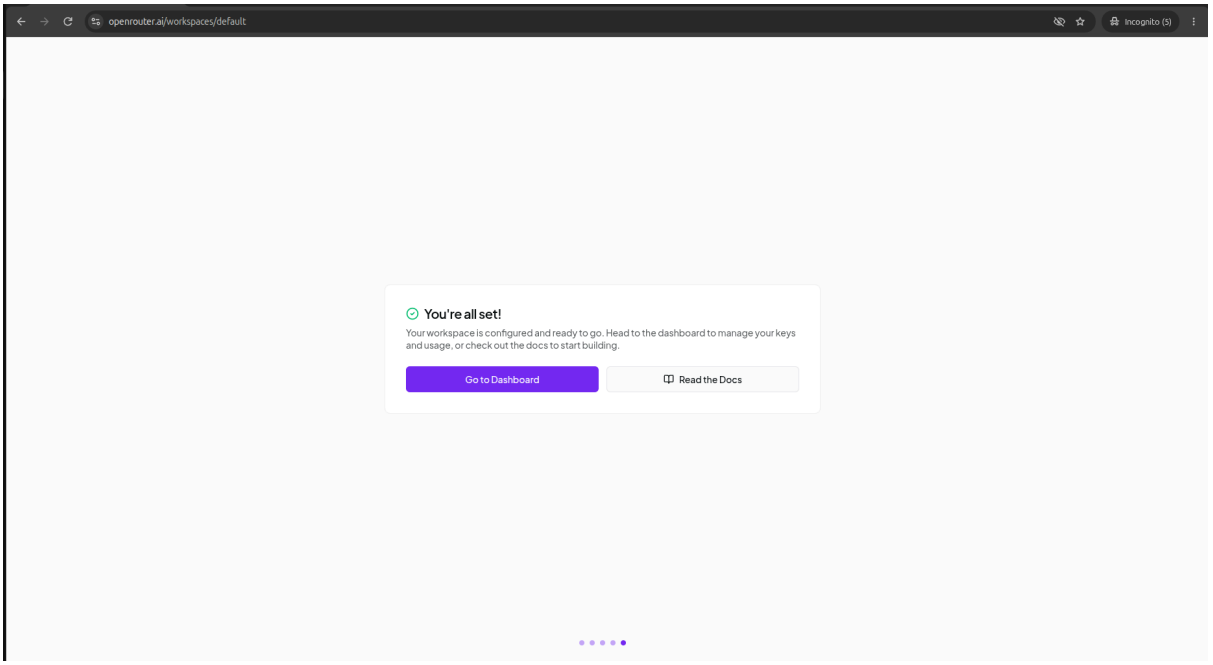

mail.google.com/mail/u/0/?service=mail&flowName=GlifWebSignIn&flowEntry=AccountChooser&ec=asw-gmail-globalnav-signin#inbox/FMfcgzQhWxRsCLZTNJKrlQbWPVNrCJfg
Incognito (5)
Gmail
Search mail
Upgrade
Compose
Inbox 868
Starred
Snoozed
Sent
Drafts 29
Purchases 2
More
Labels
1 of 1,166
Your sign up link
Inbox
OpenRouter <notifications@openrouter.ai>
to me
6:46 PM (0 minutes ago)
Sign up to OpenRouter
Click the button below to sign up to OpenRouter. This link will expire in 10 minutes.
Sign up to OpenRouter
If you're having trouble with the above button, click here.
This email link was requested from 2401:4900:1cc9:7d91:79b9:6315:9219:e782, Chennai, India at 15 October 2026, 13:16 UTC. If you didn't make this request, you can safely ignore this email.
© 2026 OpenRouter
Reply
Forward

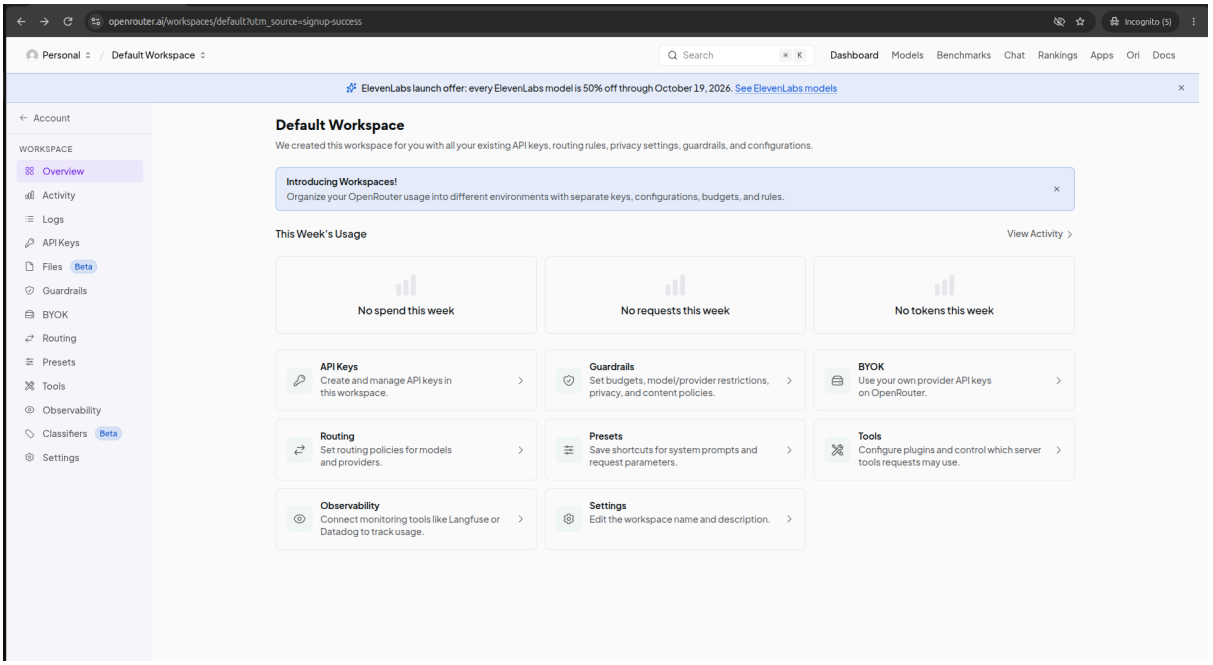

openrouter.ai/workspaces/default
Incognito (5)
Welcome to OpenRouter
The unified API for every AI model. Build with the best, switch anytime.
Unified access
GPT, Claude, Gemini, Llama, and hundreds more through one integration.
Smart routing
Automatic fallbacks and load balancing so your app never goes down.
Pay as you go
No subscriptions. Only pay for the tokens you actually use.
How will you be using OpenRouter?
You can change this later.
Individual
Build side projects, explore models, and prototype ideas
Organization
Collaborate with your team and manage usage centrally
Next

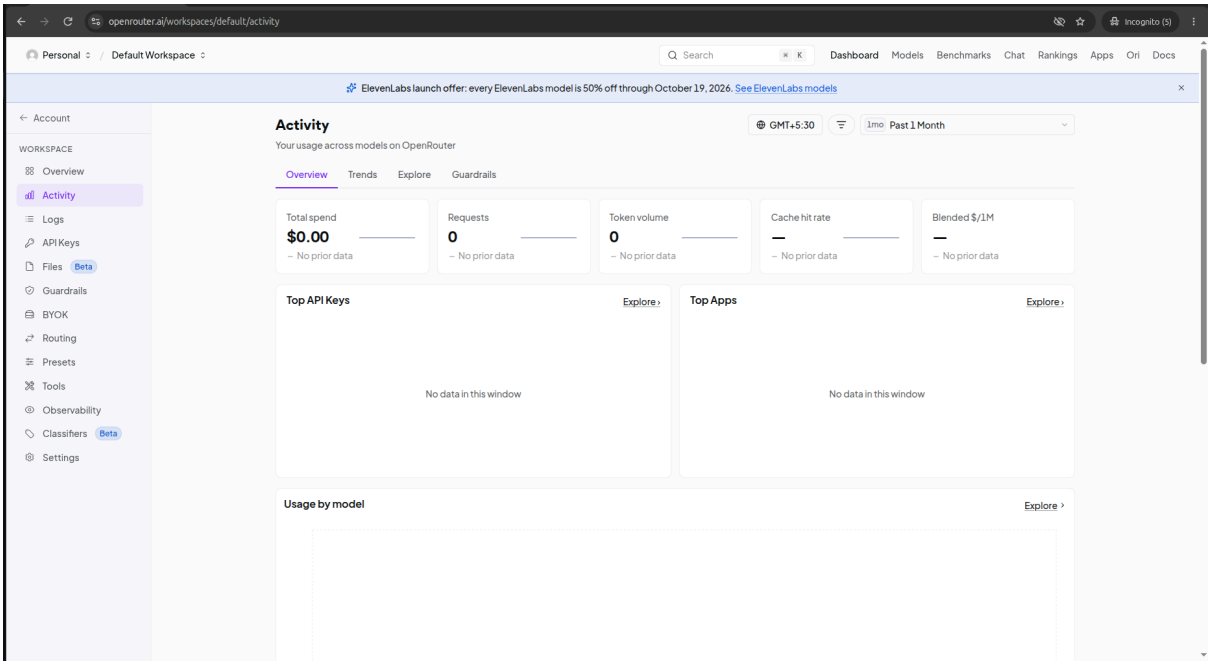

openrouter.ai/workspaces/default
Incognito (5)
Your workspace is ready
A workspace keeps your API keys, usage tracking, and billing in one place. We've created your first one with an API key so you can start making requests right away.
Your API Key
sk-or-v1-a9f·····b519
This is the only time your full key will be shown. You can always generate a new one later.
To limit the damage if this key leaks, it has a $100 credit limit and expires on Apr 13, 2027. Once it has used that much, requests are rejected until you raise the limit on your Keys page. To change the expiration, create a new key.
Make your first request
TypeScript
REST API
Copy
fetch('https://openrouter.ai/api/v1/chat/completions', {
method: 'POST',
headers: {
Authorization: 'Bearer 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',
'HTTP-Referer': '<YOUR_SITE_URL>',
'X-Title': '<YOUR_SITE_NAME>',
'Content-Type': 'application/json',
},
← Back
Continue


openrouter.ai/workspaces/default
Incognito (5)
Add a payment method
Set up your payment method — no charge yet.
Full name
Pari Margu
Country or region
United Kingdom
Address line 1
A billing address is required to verify your identity and help prevent fraud.
Complete address details to continue
← Back
I'll do this later


openrouter.ai/workspaces/default
Incognito (5)
Where did you first hear about OpenRouter?
This helps us understand how people find us.
GitHub
Friend / Colleague
Newsletter
Reddit
Conference
Billboard or transit ad
Podcast
ChatGPT / Perplexity / Claude
LinkedIn
Google
X / Twitter
YouTube
Other / Not sure
← Back
Continue


openrouter.ai/workspaces/default
Incognito (5)
You're all set!
Your workspace is configured and ready to go. Head to the dashboard to manage your keys and usage, or check out the docs to start building.
Go to Dashboard
Read the Docs


openrouter.ai/workspaces/default?utm_source=signup-success
Incognito (5)
Personal
Default Workspace
Search
Dashboard
Models
Benchmarks
Chat
Rankings
Apps
Ori
Docs
ElevenLabs launch offer: every ElevenLabs model is 50% off through October 19, 2026. See ElevenLabs models
Account
WORKSPACE
Overview
Activity
Logs
API Keys
Files
Beta
Guardrails
BYOK
Routing
Presets
Tools
Observability
Classifiers
Beta
Settings
Default Workspace
We created this workspace for you with all your existing API keys, routing rules, privacy settings, guardrails, and configurations.
Introducing Workspaces!
Organize your OpenRouter usage into different environments with separate keys, configurations, budgets, and rules.
This Week's Usage
View Activity
No spend this week
No requests this week
No tokens this week
API Keys
Create and manage API keys in this workspace.
Guardrails
Set budgets, model/provider restrictions, privacy, and content policies.
BYOK
Use your own provider API keys on OpenRouter.
Routing
Set routing policies for models and providers.
Presets
Save shortcuts for system prompts and request parameters.
Tools
Configure plugins and control which server tools requests may use.
Observability
Connect monitoring tools like Langfuse or Datadog to track usage.
Settings
Edit the workspace name and description.




Refer the code Jupyter Notebook file(.ipynb)

Many modern Large Language Models (LLMs) use a neural network architecture called the Transformer.

**1. First, what is an LLM?**

LLM stands for Large Language Model.
- Large: It may contain millions or billions of learned numbers called parameters.
- Language: It works with text, such as questions, stories, and conversations.
- Model: It is a mathematical system trained to recognize patterns and generate language.

Examples of LLMs include GPT, DeepSeek, Llama, and Qwen.

Real-world analogy: A student who has read many books
Imagine a student who has read thousands of books, stories, science articles, and conversations.

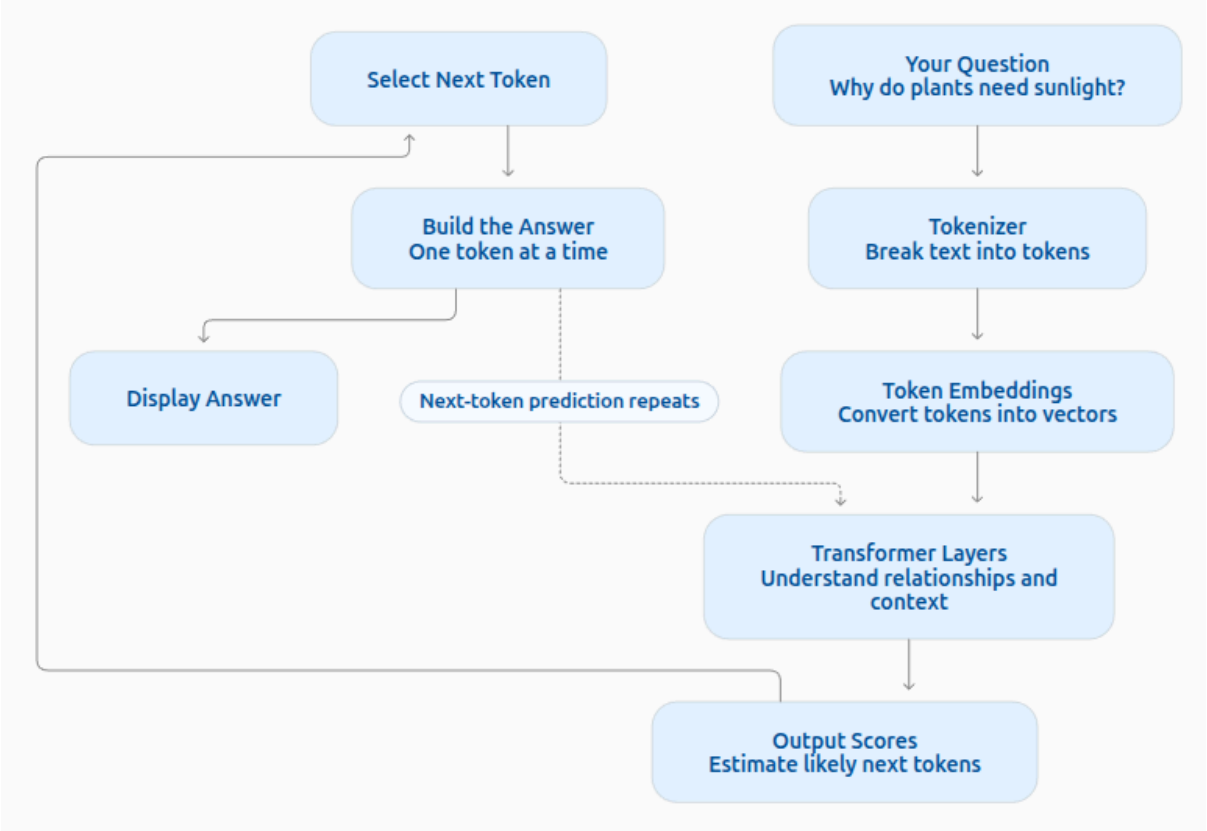
Over time, the student learns patterns such as:
- After “The sun rises in the…”, the word “east” is likely.
- After “How are you?”, a natural reply might be “I am fine.”
- If someone asks “What is 5 + 5?”, the answer is 10.

An LLM learns statistical patterns from training data. Unlike a human student, it does not learn through human experiences or understand everything in the same way people do.
The big idea: An LLM uses learned patterns to predict and generate the next piece of text.

## 2. The big picture: How an LLM answers a question

Suppose you ask:
Why do plants need sunlight?

Your question travels through several stages before the model produces an answer.




Think of it like a school examination:

| LLM stage | School analogy |
|---|---|
| Tokenizer | Breaking a sentence into words or word pieces |
| Embeddings | Giving each token a numerical identity and meaning-related representation |
| Transformer | Looking at the question and connecting related ideas |
| Output scores | Considering possible next words |
| Token selection | Choosing the next piece of text |
| Generation loop | Writing the answer one piece at a time |

These stages describe a simplified text-generation path. Real models can use more complex processing, and their exact architecture depends on the model.

The **three major Transformer architecture types** are encoder-only, decoder-only, and encoder-decoder.

**1. Encoder-only**

Reads the whole input at once and builds a rich understanding of it. Each token can look at tokens both before and after it (bidirectional attention), so the model captures full context. It doesn't generate new text; it produces representations used for understanding tasks.

Best for: classification, sentiment analysis, named entity recognition, search/embeddings
Examples: BERT, RoBERTa, DistilBERT

**2. Decoder-only**

Generates text one token at a time, left to right. Each token can only look at the tokens before it (causal attention), which makes it a natural fit for predicting "what comes next." This is the architecture behind most modern chatbots.

Best for: text generation, chat, code generation, reasoning
Examples: GPT series, Claude, Llama, Mistral

**3. Encoder-decoder (sequence-to-sequence)**

Combines both. The encoder reads and understands the input, then the decoder generates the output while attending to the encoder's representation (cross-attention). This suits tasks where you transform one sequence into another.

Best for: translation, summarization, question answering
Examples: the original Transformer, T5, BART

**Quick comparison**

| Type | Attention | Main strength | Example |
|---|---|---|---|
| Encoder-only | Bidirectional | Understanding text | BERT |
| Decoder-only | Causal (left-to-right) | Generating text | GPT, Llama |
| Encoder-decoder | Both + cross-attention | Transforming text | T5 |

**Simple analogy:** the encoder is like a reader who fully comprehends a passage, the decoder is like a writer who produces text word by word, and the encoder-decoder is a translator who reads in one language and writes in another.

# Prompt Techniques

Please refer the document: prompting_techniques.md

# Difference between Open-Weight Models vs Open-Source Models (True Open Source)

The main difference is that open-weight models share only the trained model parameters (the "brain"), whereas open-source models share the weights, code, and complete training blueprints or data needed to fully recreate and audit the system.

**Open-Weight Models**

• **What is released:** Only the trained neural network parameters (numerical weights) and often the runtime/inference code.
• **What is hidden:** The training dataset, pre-processing methods, or full training pipeline remain a trade secret.
• **Capabilities:** You can run them locally on your own hardware, fine-tune them, or build applications on top of them.
• **Limitations:** You cannot independently verify training data for bias, safety, or copyright issues because the ingredients are missing.
• **Examples:** Meta's Llama series, Mistral, and DeepSeek.

**Open-Source Models (True Open Source)**

• **What is released:** Complete transparency—including weights, architecture code, training scripts, documentation, and training data information (meeting standards like the Open Source Initiative AI definition).
• **Capabilities:** You can inspect, modify, audit, re-train from scratch, and share every single layer of the AI system.
• **Limitations:** They are rarer in large frontier scales due to the massive cost and copyright sensitivities of releasing raw training datasets.
• **Examples:** OLMo (by the Allen Institute) and BLOOM.

| Feature | Open-Weight | Open-Source (True) |
|---|---|---|
| **Trained Weights** | Yes | Yes |
| **Inference/Model Code** | Usually Yes | Yes |
| **Training Code / Scripts** | Often No | Yes |
| **Training Data Transparency** | No (Secret) | Yes / Public |
| **Local Execution & Fine-Tuning** | Yes | Yes |
| **Full Auditability & Replicability** | No | Yes |

# Types of Language Models(LMs)

Language models (LMs) in Generative AI are specialized deep learning systems designed to understand, process, and generate human-like text or multimodal content. They are generally categorized based on their architecture, scale, access control, and functional purpose.

**1. By Core Architecture & Functionality**

• Generative Pretrained Transformers (GPT): General-purpose text understanding and generation models trained via next-token prediction on massive corpora (e.g., OpenAI's GPT-4, GPT-3).
• Mixture of Experts (MoE): Models that utilize massive total parameter counts but selectively activate only a subset of pathways ("experts") per token to scale capacity and improve efficiency (e.g., Mixtral, DeepSeek-V3/R1).
• Large Reasoning Models (LRMs): Models specifically tuned for multi-step logical deduction, chain-of-thought processing, math, coding, and tool integration (e.g., DeepSeek-R1, OpenAI o-series).
• Vision-Language Models (VLMs): Multimodal models trained to process and ground both text and image/video inputs simultaneously (e.g., multimodal GPT-4, Gemma-3).
• Large Action Models (LAMs): Agentic models built to plan, interact with software, use APIs, or drive digital/robotic actions autonomously.
• Text-to-Text / Encoder-Decoder Models: Architecture variants (such as Google's T5) that treat both inputs and outputs as text strings, streamlining translation, summarization, and task adaptation.

**2. By Model Size & Deployment Footprint**

• Large Language Models (LLMs): Massive models with billions to trillions of parameters requiring heavy cloud infrastructure, capable of open-ended dialogue and complex cross-domain tasks (e.g., GPT-4, Claude 3.5, Llama 3/4).
• Small Language Models (SLMs): Compact, lightweight models (typically 1B to 14B parameters) designed for fast inference, low energy use, and local or on-device edge execution on phones and laptops (e.g., Microsoft's Phi-4, SmolLM2, Gemma-4B).

**3. By Access & Control (Open vs. Proprietary)**

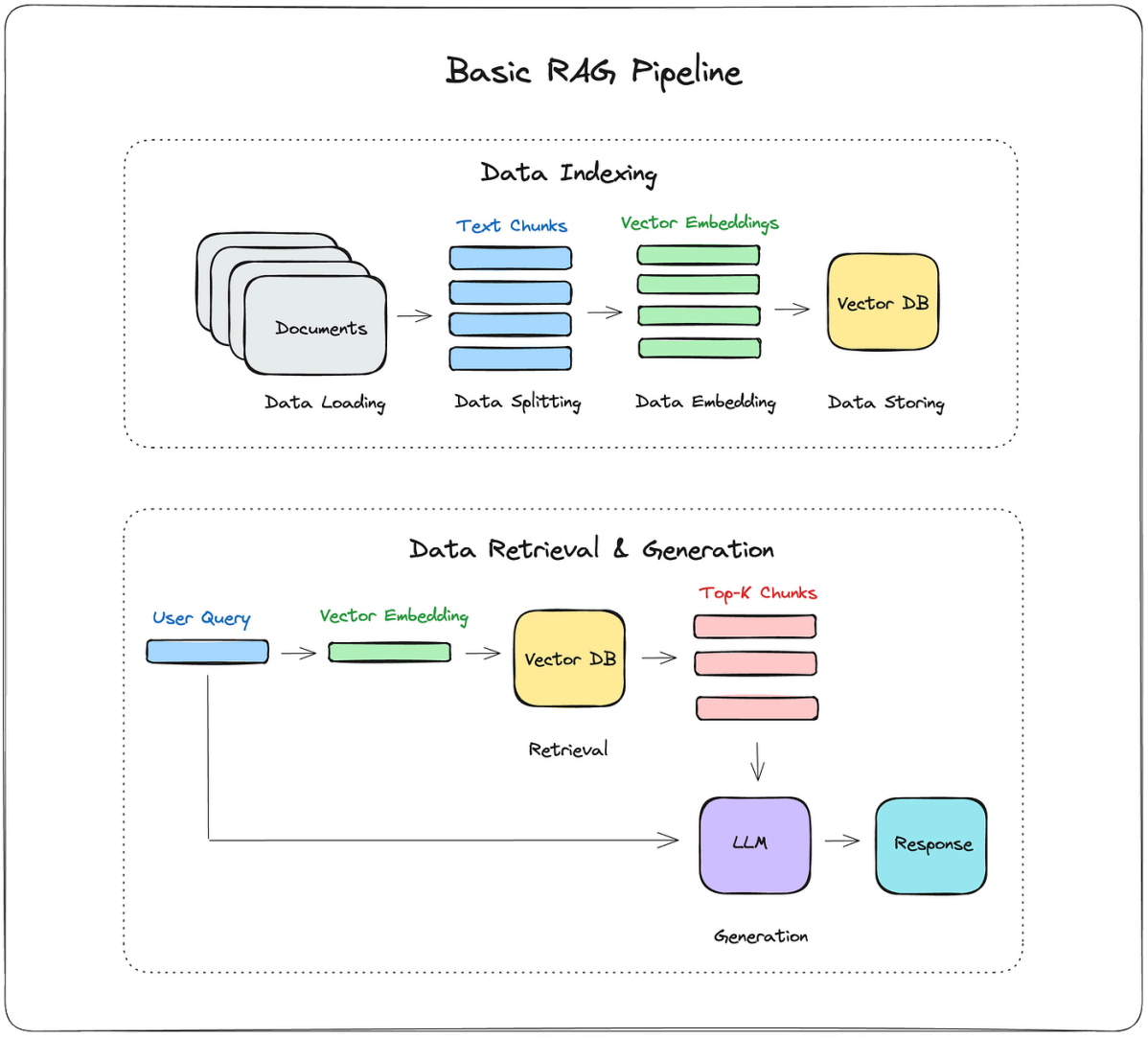
• Proprietary / Closed-Source Models: Commercial models hosted via cloud APIs where internal weights and architectures are hidden, managed by providers like OpenAI, Anthropic, and Google.
• Open-Source / Open-Weight Models: Models with publicly released weights, code, or training recipes that can be audited, self-hosted, or fine-tuned locally (e.g., Meta's Llama series, Mistral, DeepSeek).

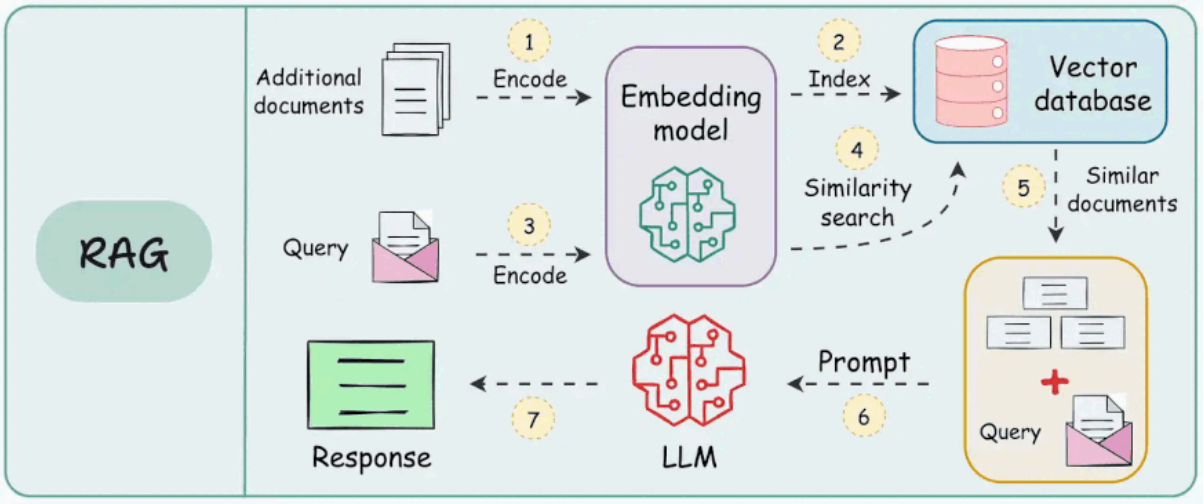
**4. By Specialization & Domain Scope**

• General-Purpose Models: Jack-of-all-trades models built for broad conversational, creative, and analytical versatility.
• Domain-Specific / Specialized Models: Fine-tuned or custom-trained on vertical data for specific fields like medicine (BioBERT), legal documents (Legal BERT), or software engineering (Codestral).

# Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI framework that improves large language models (LLMs) by connecting them to external knowledge bases before generating a response.



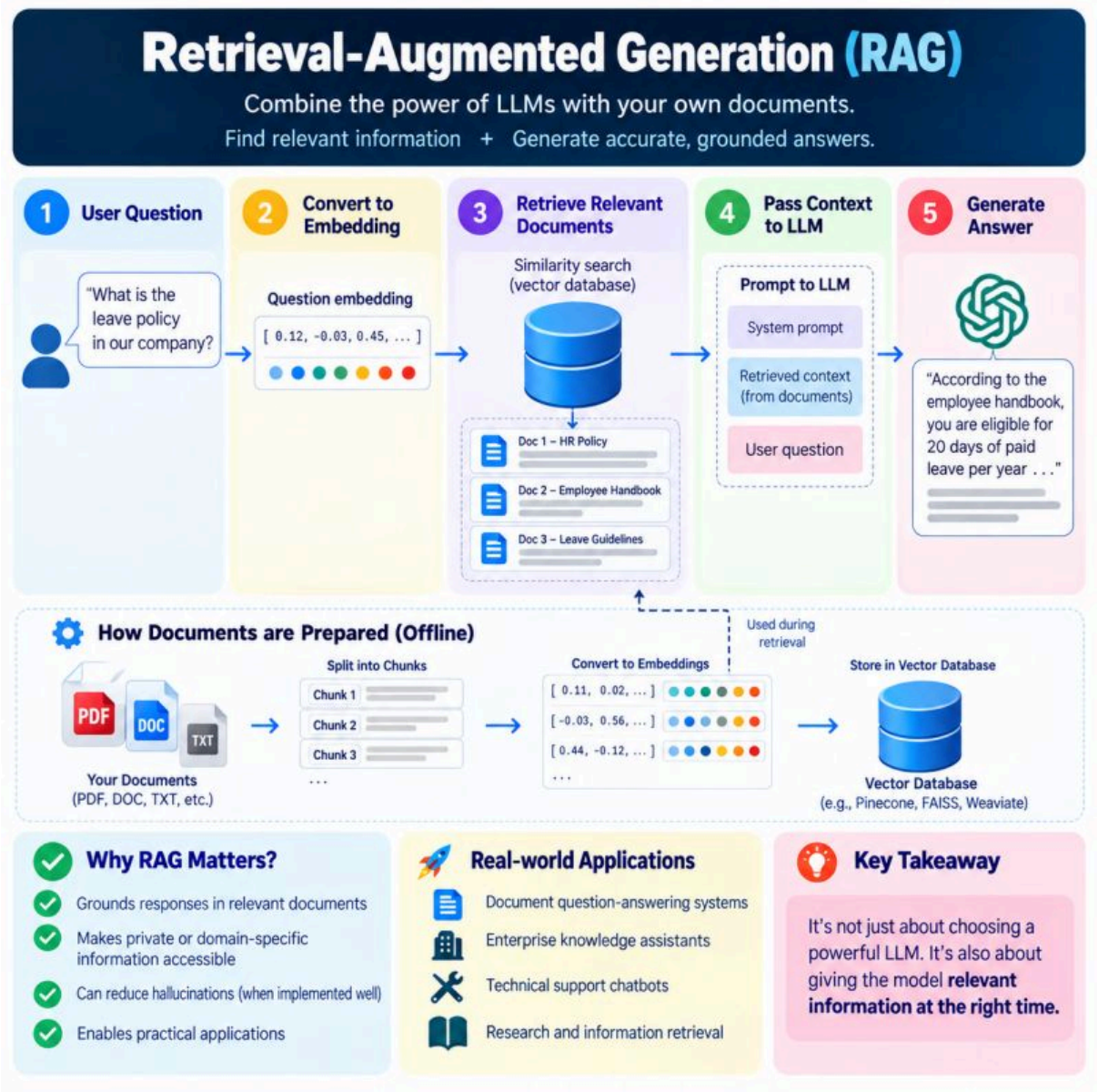



How RAG Works

1. Retrieval: When a user asks a question, the system searches external data sources (such as company documents, databases, or web pages) for relevant information.

2. Augmentation: The retrieved data is added to the user's original prompt as context.

3. Generation: The LLM uses this custom context to synthesize an accurate, grounded answer.




Key Benefits

• Reduces Hallucinations: Grounds answers in verified, factual sources rather than relying purely on pre-trained memory.
• Keeps Data Current: Bypasses the static cutoff dates of LLMs by pulling real-time or updated information dynamically.
• Saves Cost and Time: Avoids the need for expensive, time-consuming model retraining or fine-tuning.
• Domain Specialization: Tailors general AI models to private enterprise data, scholarly articles, or specific codebases.

# Introduction to Generative AI(GenAI)

## What is AI?

**Artificial Intelligence (AI)** is a technology that enables computers and machines to perform tasks that typically require human intelligence, such as learning, understanding language, recognizing images, solving problems, and making decisions.

In simple words, AI makes machines capable of performing tasks that normally require human thinking.

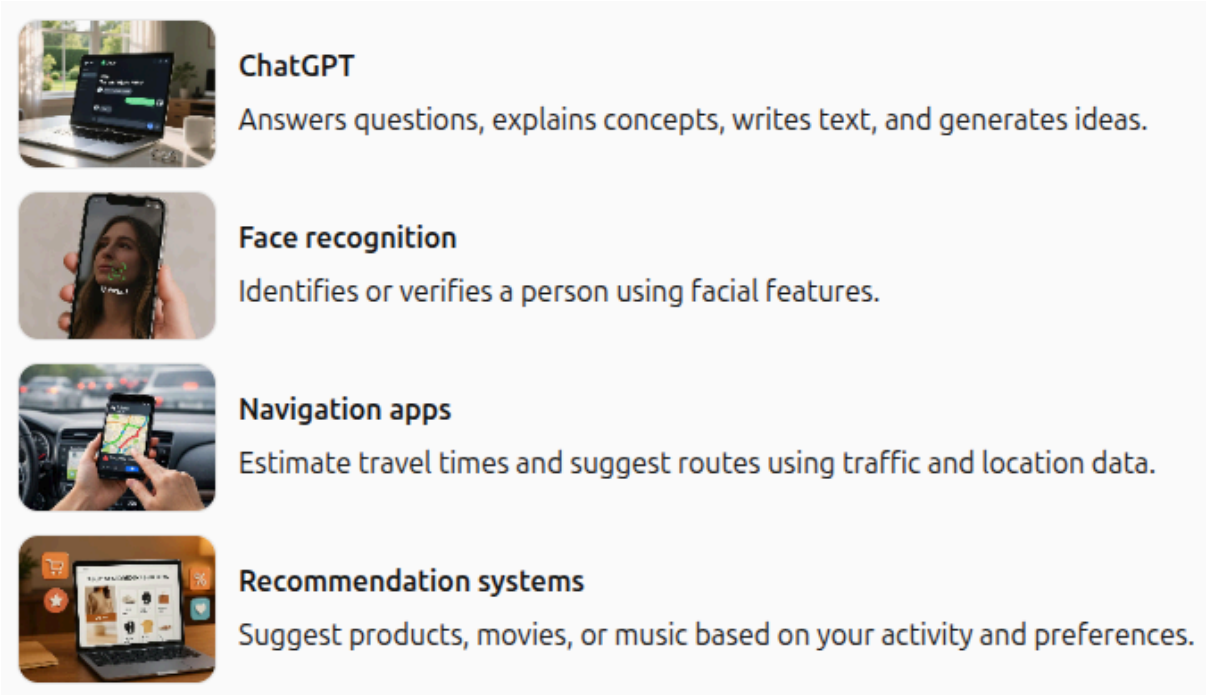
How does AI work?

**1. Input data**
AI receives information such as text, images, audio, or numbers.

**2. Processing and learning**
AI uses algorithms and, in many cases, patterns learned from data to analyze information and produce a result.

**3. Output**
AI generates a response, prediction, recommendation, or action based on the task.

Examples of AI in everyday life



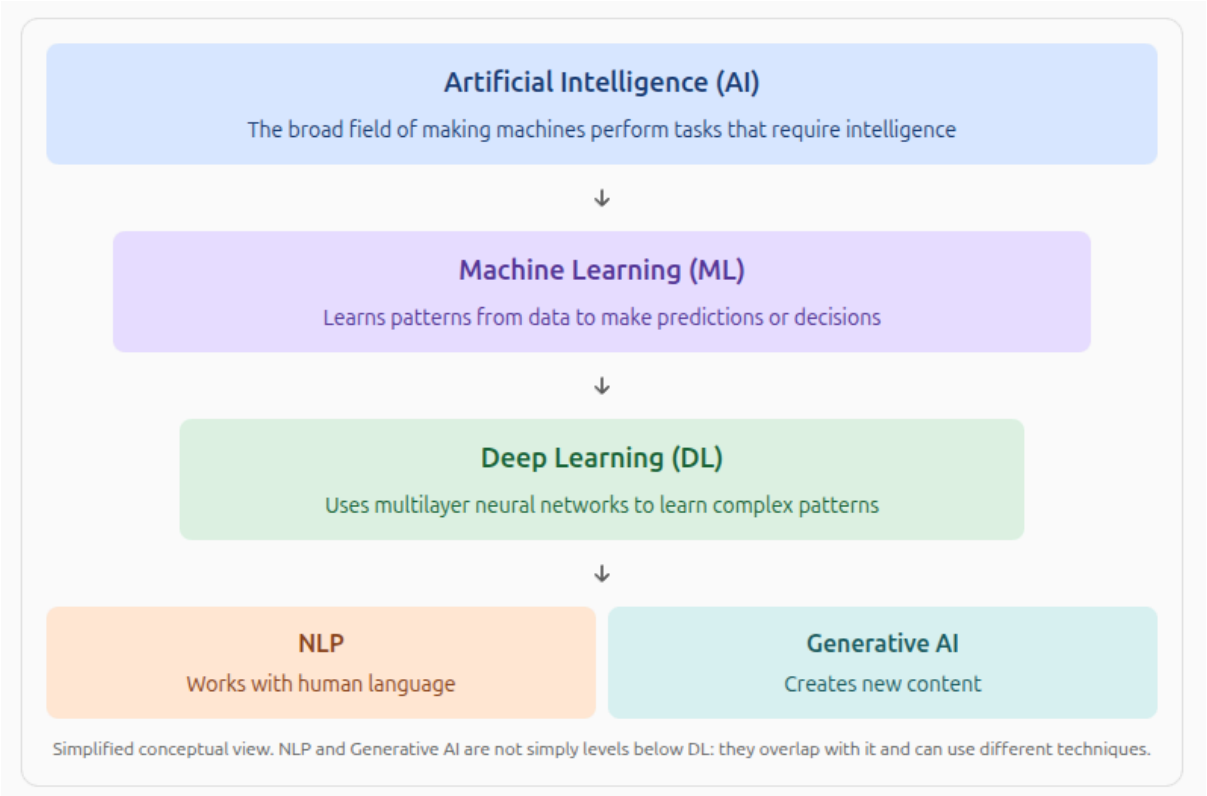

Types of AI

| Type | Description | Example |
|---|---|---|
| Narrow AI | Designed for specific tasks | Voice assistants, spam filters |
| Generative AI | Creates new content | ChatGPT, image generators |
| General AI (AGI) | Human-level capability across a broad range of tasks | A hypothetical form of AI; not yet achieved |

What is the relationship between AI, ML, DL, NLP, and Generative AI?

AI, ML, DL, NLP, and Generative AI are closely related fields, but they have different roles. The easiest way to understand them is to see AI as the broad field, ML as a way to achieve AI, DL as a type of ML, and Generative AI as a field focused on creating new content. NLP focuses on working with human language.

1. Relationship between AI, ML, DL, NLP, and Generative AI




The key distinction is that AI, ML, and DL describe increasingly specific approaches to intelligent systems, while NLP describes a language-focused area and Generative AI describes systems that generate content. Modern NLP and Generative AI often use deep learning.

**For example:** ChatGPT is a generative AI system. You provide a prompt, the model processes it, and it generates a response in natural language.

One important point: AI does not necessarily think or understand the world in the same way humans do. Its capabilities and limitations depend on how it is designed and trained.

2. What does each term mean?

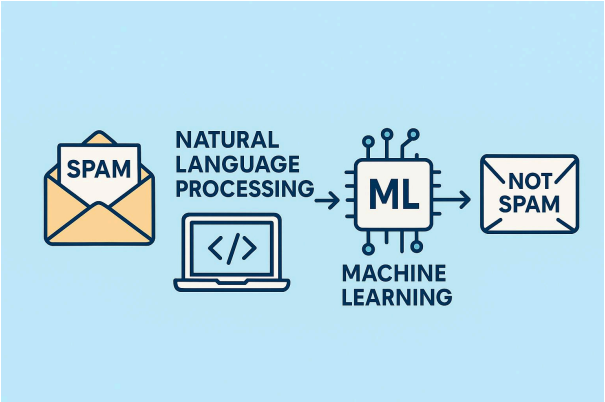
| Term | Full form | Main purpose | Example |
|---|---|---|---|
| AI | Artificial Intelligence | Makes machines perform intelligent tasks | Chess-playing system |
| ML | Machine Learning | Learns patterns from data | Predicting house prices |
| DL | Deep Learning | Learns complex patterns using neural networks | Image recognition |
| NLP | Natural Language Processing | Processes and understands human language | Language translation |
| Generative AI | Generative Artificial Intelligence | Generates new content | ChatGPT generating an answer |

3. How are they connected in real-world applications?

Example 1: Spam email detection

- AI: the overall intelligent application.
- ML: learns patterns from spam and legitimate emails.
- NLP: may analyze the email's text.
- DL: may be used to identify complex patterns.

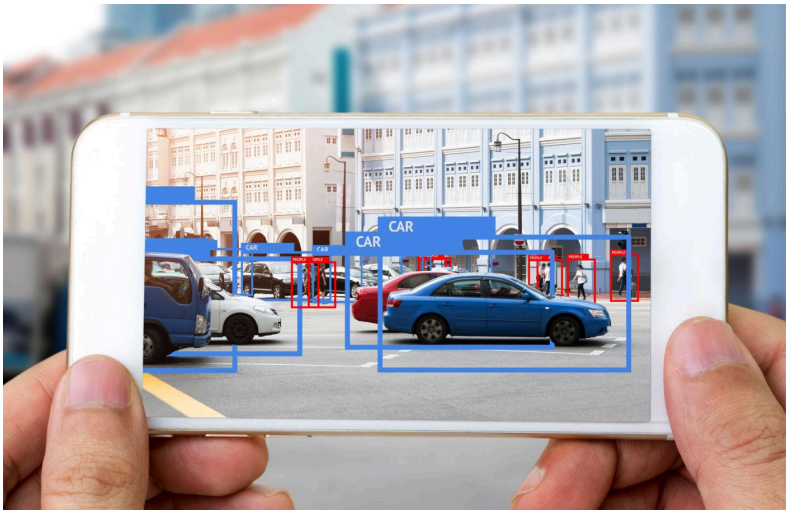
The system classifies an email as spam or legitimate; it does not need Generative AI.




Example 2: Image recognition

- AI: the overall application.
- ML: learns from labeled image data.
- DL: uses neural networks to recognize objects.

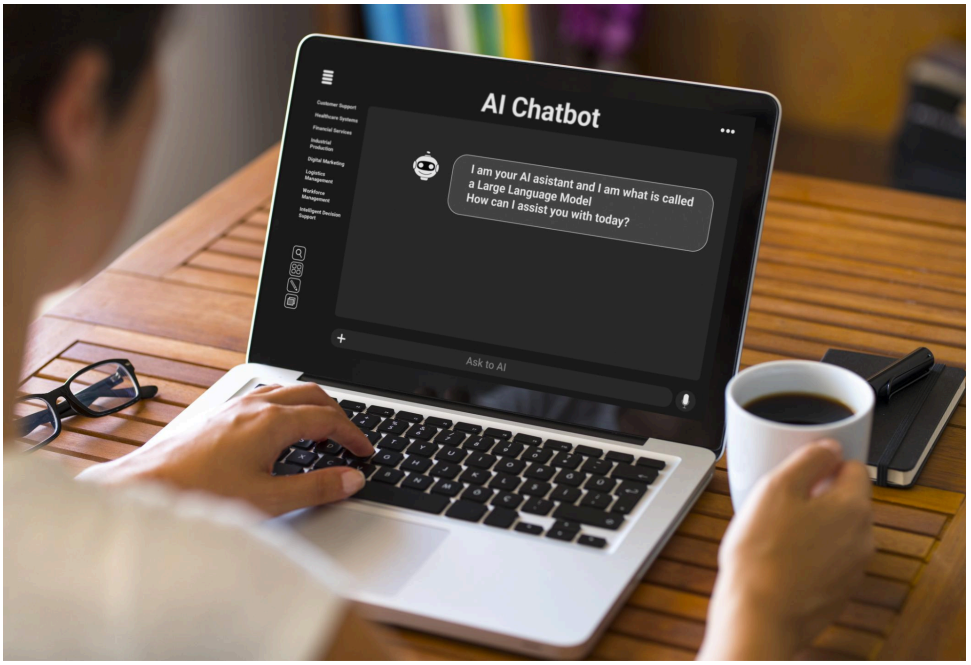
NLP is not essential because the primary input is an image.




Example 3: ChatGPT

- AI: the overall intelligent system.
- ML: learns from training data.
- DL: uses large neural networks called transformers.
- NLP: supports processing and generating human language.
- Generative AI: generates new responses, summaries, code, and other content.

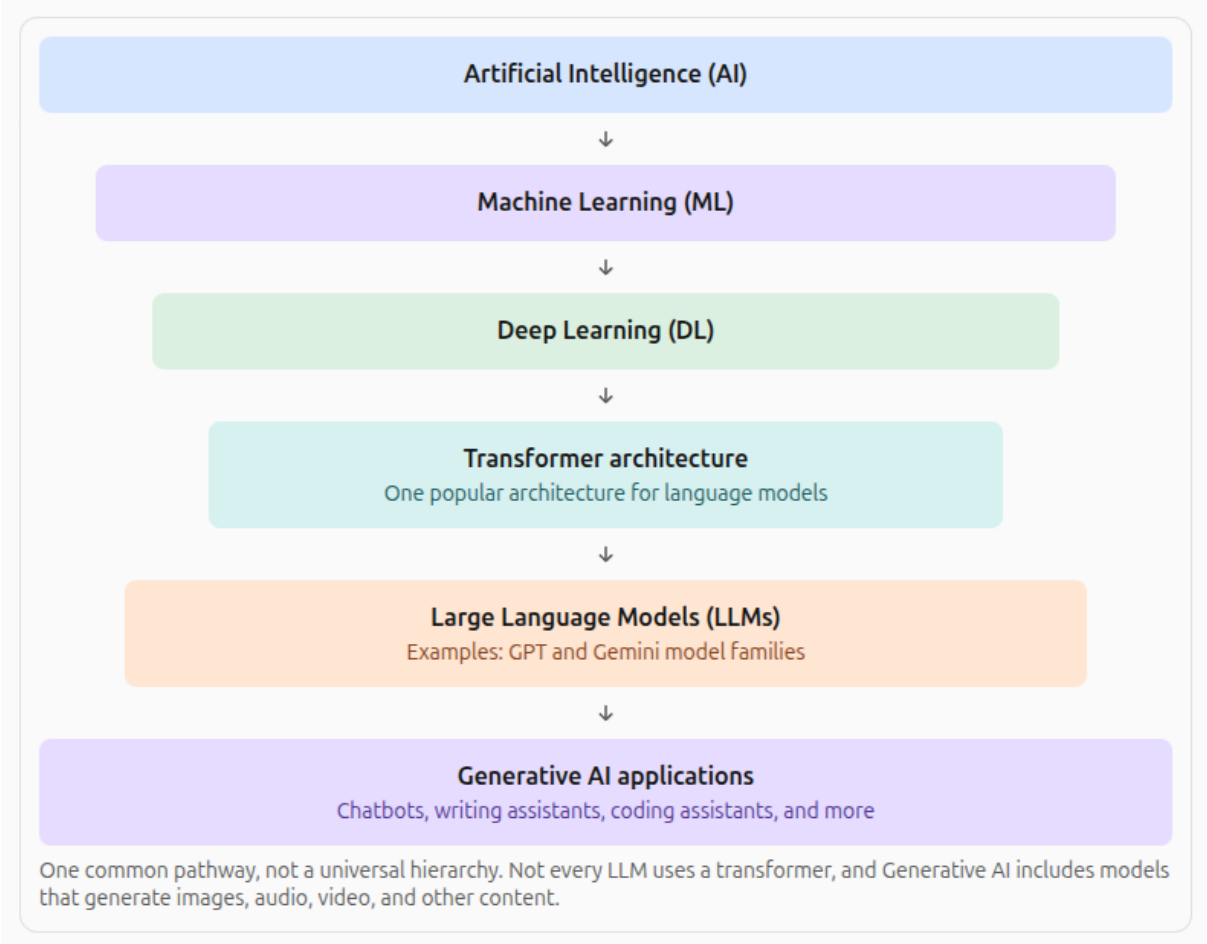
ChatGPT brings several of these concepts together in one application.




4. Where do LLMs fit in?

LLM stands for Large Language Model.

An LLM is a type of model, typically based on deep learning, trained on large amounts of data to process and generate language. LLMs power many Generative AI applications.




5. Remember the difference with this simple example

Imagine building a system that answers questions about cricket.

- AI: The overall system designed to answer questions intelligently.
- ML: The system learns patterns from training data.
- DL: A neural network learns complex language patterns.
- NLP: The system processes the user's question and works with language.
- Generative AI: The model generates a new answer in natural language.
- LLM: The large language model that may power that answer.

In one sentence: AI is the broad field, ML learns from data, DL uses deep neural networks, NLP works with human language, and Generative AI creates new content. These concepts overlap rather than forming five completely separate technologies.

## What is Generative AI?

Generative AI is a type of artificial intelligence that **creates new content**, such as text, images, audio, video or code. Most traditional AI only analyzes or classifies existing data.

It works by learning patterns from huge amounts of training data. When you give it a prompt, it uses those patterns to produce something new that wasn't stored anywhere before. A large language model (LLM) like GPT or Gemini, for example, writes a response one word at a time, each time predicting the most likely next word based on your prompt and everything it has written so far.

This is the same flow as your diagram: **user prompt → generative AI model → generated new content → response to user**, with ChatGPT as the chat interface around it.

**Common examples:**

- **Text:** ChatGPT, Gemini, Claude write answers, essays, emails and summaries.
- **Images:** DALL·E, Midjourney, Stable Diffusion create pictures from text descriptions.
- **Code:** GitHub Copilot suggests and writes programming code.
- **Audio and video:** some tools generate voices, music or short video clips.

# Introduction to Retrieval-Augmented Generation (RAG)

RAG is

**References:**

# LLM TOOLS CHEATSHEET




## General-Purpose LLM Providers

Used for training, fine-tuning, & deploying LLMs

- Claude
- OpenAI
- Google Gemini
- LLaMa by Meta
- MISTRAL AI_
- Grok
- deepseek
- cohere
- AI21labs

## AI Agent & Multi-Agent Systems

Orchestrate tasks, agents, memory, and planning

- AutoGen
- crewai
- Superagent
- CAMEL-AI
- MGX
- manus
- AgentVerse
- LangGraph
- AUTOGPT

## LLM Application Frameworks

Build full-stack LLM apps with memory, chains, and tools

- LangChain
- LlamaIndex
- Haystack by deepset
- OpenLLM
- kernel
- DSPy
- Guardrails AI

## Prompt Engineering Tools

Craft, store, evaluate, and manage prompts

- PromptLayer
- Promptable
- Langfuse
- ChainForge
- PromptPerfect
- PromptTools

## Vector Databases

Store embeddings for RAG & semantic search

- FAISS
- ChromaDB
- Pinecone
- Weaviate
- redis
- Qdrant
- Milvus

## No-Code / Low-Code AI Builders

Build LLM apps without writing heavy code

- Flowise
- Langflow
- n8n
- replit
- CURSOR
- Lovable
- Chatbase
- Relevance AI
- V0